

CS-670 UNDERGRADUATE PROJECT REPORT

Privacy, Representation, and Noise Propagation in Federated Recommendation

A Reproducible Study of Low-Rank Models, Local Personalization, and User-Level
Differential Privacy

Final project report

5 October 2026

Evidence coverage: Phase 0; baselines B0–B4; controlled extension E1; noise-memory studies E2/E2b; MovieLens-1M replication E3; and ranking-tail screen E4. Experiments completed through 3 October 2026.

Abstract

Federated recommendation separates shared item learning from local user personalization, but useful recommendations can deteriorate when whole-user differential privacy (DP) requires clipping and noisy aggregation. This project studies whether a smaller shared representation improves that trade-off and, more broadly, how injected noise becomes disturbance to item matrices, personalized scores, and rankings. A reproducible MovieLens-100K pipeline compares centralized Bayesian Personalized Ranking (BPR), a user-weighted centralized control, and federated full-matrix and low-rank models with and without DP. Five-seed historical comparisons find low-rank private models better than the private full-matrix reference in four of sixteen multiplicity-adjusted comparisons; all four nonprivate low-rank variants underperform the converged full model. These results motivate a separately frozen, equal-budget comparison of full, jointly trained two-factor, and fixed-public-factor parameterizations.

The fixed-factor extension obtains no positive adjusted primary comparison and loses eight of sixteen comparisons to joint factor training. A user-level private popularity baseline exceeds every collaborative point estimate at target $\epsilon \leq 4$ in that extension. Conditional moment calculations and measured perturbations show why factor-coordinate noise norms are inadequate proxies for recommendation damage: effective noise depends on factor geometry, local user scale, clipping, and pairwise margins. Pulse-and-reset studies initially reveal large two-factor trajectory differences, but an orthogonal alignment control substantially reduces them. Thirty transferred MovieLens-1M training runs and 200 further probes reproduce the dependence on the way noise is coupled across rebalanced factor trajectories. A separate screen of 3,200 item-pair records finds a maximum absolute sign-flip probability error of 7.67×10^{-6} for a variance-matched Gaussian approximation and rejects non-Gaussian tails as a practically consequential explanation in the sampled states.

The completed contribution is an audited experimental study and a more precise interpretation of noise diagnostics in personalized low-rank recommendation. It does not establish a new state-of-the-art algorithm or a confirmed claim of paper-level novelty. The report includes the full methodological progression, unfavorable findings, privacy scope, statistical qualifications, mathematical derivations, and a separately generated supplementary results document.

Keywords: federated recommendation; user-level differential privacy; matrix factorization; BPR; effective noise; personalization; factor alignment; reproducible evaluation.

Scope of this submission

The main report explains the completed research and includes essential tables and all nine curated figures. The companion *Supplementary Results and Experimental Record* provides complete retained utility tables, primary and secondary contrasts, diagnostic summaries, and source mappings. Numerical results come from frozen saved outputs; preparation of this report did not retrain models or score additional test targets. MovieLens-100K is a repeatedly used development benchmark. MovieLens-1M validation has been evaluated; its test targets remain unscored.

Contents

Abstract	i
Notation and reading conventions	vi
1 Introduction	1
1.1 Problem and motivation	1
1.2 Research questions	1
1.3 Completed contributions	2
1.4 Organization	2
2 Related work and contribution boundaries	3
2.1 Ranking, federation, and private collaborative learning	3
2.2 Personalization and public information	3
2.3 Low-rank geometry and fixed-factor methods	3
2.4 Noise propagation and ranking stability	4
2.5 How the literature was used	4
3 Problem formulation and privacy model	4
3.1 Personalized implicit-feedback ranking	4
3.2 Protected information and adjacency	5
3.3 Private aggregation mechanism	5
3.4 Release scope and local personalization	6
4 Datasets and evaluation protocol	6
4.1 MovieLens-100K and implicit conversion	6
4.2 Negative sampling and candidate filtering	7
4.3 Ranking metrics	8
4.4 MovieLens-1M replication	8
4.5 Holdout exposure and dataset limitations	8
5 Models and implementation	8
5.1 B0 and B0-UW: centralized controls	8
5.2 B1: federated full matrix without DP	9
5.3 B2: private full matrix	9
5.4 B3 and B4: jointly trained item factors	9
5.5 E1: a separately controlled comparison	10
5.6 User-level private popularity	10
5.7 Communication accounting	11
6 Experimental design and statistical analysis	11
6.1 Sequential controls and preserved stages	11

6.2	Seeds and paired uncertainty	12
6.3	Rank selection, groups, and personalization diagnostics	12
6.4	Diagnostic replication and advancement criteria	12
7	Results of the historical baselines	13
7.1	Sanity references and centralized learning	13
7.2	Nonprivate low rank	14
7.3	Private full matrix and low rank	14
7.4	Communication and the original hypothesis	15
8	E1: effective noise and fixed-factor training	16
8.1	Why the effective matrix matters	16
8.2	Controlled utility comparison	16
8.3	The private popularity baseline changes the reference point	17
8.4	Measured geometry, clipping, and numerical behavior	18
8.5	Activity groups and personalization ablations	19
8.6	Recommendation-specific implications	19
9	E2/E2b: local-state memory and noise coupling	19
9.1	Question and intervention design	19
9.2	Score decomposition and measurement	20
9.3	Initial findings and why a control was needed	20
9.4	Orthogonal alignment control	20
9.5	Verification and limits	21
10	E3: larger-data replication and balancing controls	22
10.1	Transferred training and factorial diagnostic	22
10.2	Validation utility	23
10.3	Main diagnostic result	23
10.4	Checks and research decision	24
11	E4: a conditional ranking-tail screen	26
11.1	Candidate explanation and established mathematics	26
11.2	Synthetic checks and real-state selection	26
11.3	Outcome and numerical scope	26
12	Discussion, limitations, and conclusions	27
12.1	Answers to the research questions	27
12.2	What the combined evidence explains	28
12.3	Threats to validity	28
12.4	Future work justified by the evidence	29
12.5	Conclusion	29

References	31
A Mathematical derivations and diagnostic identities	34
A.1 Conditional effective-matrix moments	34
A.2 Score moments and pairwise margins	34
A.3 Scale changes, balancing, and clipping	34
A.4 FixedB score capacity and regularization	35
A.5 A product-preserving coupling null example	35
A.6 Exact conditional margin law and numerical evaluation	35
A.7 Analytic Gaussian calibration for popularity	36
B Implementation details and algorithm record	36
B.1 Client update and shared aggregation	36
B.2 Final parameter register	37
B.3 Diagnostic random streams and identity checks	37
B.4 Numerical environment and reproducibility precision	38
B.5 Communication ledger	38
C Material revisions and hypothesis decisions	39
C.1 Why the project history belongs in the report	39
C.2 Final hypothesis ledger	40
C.3 What was deliberately excluded from the submission tables	41
D Artifact provenance and reproduction guide	41
D.1 Separation of evidence, report source, and generated output	41
D.2 Data fingerprints	41
D.3 Evidence checks	42
D.4 Rebuilding the submission	42
D.5 Supplementary material and source map	42

List of Figures

1	Nonprivate controls and low-rank utility	14
2	Historical private utility	15
3	Controlled E1 utility	17
4	Effective noise, score scale, and clipping	18
5	Pulse/reset trajectories	21
6	The factor-coordinate coupling control	22
7	Transferred ML-1M validation utility	23
8	Replication with and without continuation balancing	25
9	The rejected ranking-tail candidate	27

List of Tables

1	Final iterative privacy calibration at $q = 0.1$, $T = 50$, $\delta = 10^{-5}$	6
2	Dataset sizes under the frozen chronological implicit-feedback protocol.	7
3	Experimental stages and their evidence units.	11
4	Matched centralized and federated controls (three seeds).	13
5	Converged nonprivate federated utility (five seeds).	13
6	Historical private and matched no-DP test NDCG@10 (five-seed means).	14
7	E1 test utility with ranks selected on validation at each privacy level.	16
8	Transferred MovieLens-1M validation NDCG@10.	23
9	E3 shared-reset top-10 disagreement at lag ten (percent of set entries).	24
10	Final retained configurations and stopping rules.	37
11	Modeled dense tensor communication for retained historical methods.	38
12	Material research iterations and their consequences.	39

Notation and reading conventions

Symbol or label	Meaning
N, M	Numbers of eligible users and catalog items.
d, r	Full embedding dimension (64) and factor rank (4, 8, 16, 32).
p_u, P	Local user vector and matrix whose rows are user vectors.
q_i, Q	Effective item vector and shared effective item matrix.
A, B	Item factors, $A \in \mathbb{R}^{M \times r}$ and $B \in \mathbb{R}^{r \times d}$, with $Q = AB$.
s_{ui}, m_{uij}	Score $p_u^T q_i$ and pairwise margin $s_{ui} - s_{uj}$.
q, E, T	Independent client inclusion probability, local epochs, and federated rounds.
C, σ	Whole-user clipping threshold and Gaussian noise multiplier on the sum.
η_ℓ, η_s	Local and server learning rates.
τ	Per-coordinate noise SD after server scaling, $\eta_s \sigma C / (qN)$.
(ϵ, δ)	User-level DP parameters for one fixed-setting mechanism.
B0 / B0-UW	Centralized interaction-weighted / user-weighted references.
B1 / B2	Federated full-matrix models without / with DP.
B3 / B4	Federated two-factor low-rank models without / with DP.
Full / Two / FixedB	Separately named E1 full, two-factor, and fixed-public-factor methods.
nodp / clip	Matched no-clipping/no-noise and clipping-only E1 controls.
Seed SD	Sample standard deviation across retained training or noise seeds.
Adjusted interval	A Bonferroni-adjusted paired-user bootstrap interval for its declared family.
Top-10 churn	One minus top-10 set overlap divided by ten, averaged over observer users.

Throughout, a number reported as mean $\pm$ SD describes seed variation, not a confidence interval. Smaller ϵ means stronger privacy at fixed δ . The labels $\epsilon = 1, 2, 4, 8$ denote targets; actual accountant bounds are reported separately. E2–E4 checkpoint labels refer to the original training mechanism, not to the privacy cost of additional diagnostic operations. A “cold target” has no training-positive interaction anywhere in the retained population; it is still present in the public evaluation catalog.

1 Introduction

1.1 Problem and motivation

Recommendation systems use interaction histories to predict which items a person will find useful. Those histories can reveal preferences and behavior that users may not want to disclose. Federated learning changes where training occurs: user records remain on clients, while a server coordinates the shared model. This separation is useful, but transmitted updates can still depend strongly on an individual’s records. Differential privacy adds a formal constraint on that dependence. In this project the protected unit is an entire user, so protecting one rating while exposing the rest of the user’s history would be insufficient.

The practical difficulty is that privacy-preserving aggregation also changes the learning problem. A client update is clipped before contributing to an aggregate, and random noise is added to that aggregate. Clipping can suppress informative directions; noise can obscure weak collaborative signals. These effects interact with client sampling, local training, the representation of items, and persistent user embeddings. A useful experimental study must distinguish these mechanisms instead of treating every difference from a centralized model as a privacy cost.

The project’s starting hypothesis was that reducing the number of shared item parameters could improve the privacy–utility–communication trade-off. A full item matrix $Q \in \mathbb{R}^{M \times d}$ can be written approximately through factors $A \in \mathbb{R}^{M \times r}$ and $B \in \mathbb{R}^{r \times d}$, with $r < d$. Sharing the factors reduces the number of communicated scalars from Md to $r(M + d)$. Under isotropic Gaussian perturbation, fewer coordinates also reduce the expected squared norm of noise in those coordinates. Neither observation establishes that recommendation scores become less disturbed. The model uses the product AB , not the separate factor coordinates, and its predictions multiply that product by a local user vector whose scale can itself change during noisy training.

The investigation consequently developed in stages. The initial baselines separated centralized learning, user weighting, federation, privacy, and low rank. Their mixed outcomes motivated controlled comparisons of parameterizations and direct measurement of effective noise. Later experiments asked whether noise persists through local personalization and whether diagnostic trajectory differences depend on arbitrary factor coordinates. A final candidate examined whether non-Gaussian margin tails matter in the saved trained states. This report presents those stages as a connected study, including the hypotheses that failed.

1.2 Research questions

Five questions organize the completed work.

- RQ1. **Representation and utility.** At matched privacy targets and a fixed private training horizon, when does a low-rank shared representation outperform a full shared matrix, and what utility is lost before privacy is introduced?
- RQ2. **Parameterization and effective noise.** Does fixing one public factor improve prediction quality once initialization, tuning opportunity, clipping controls, and a private popularity reference are included?
- RQ3. **Persistent personalization.** After an injected shared-model perturbation has influenced local user states, how much prediction disturbance remains when the shared state is restored?
- RQ4. **Validity of trajectory comparisons.** How much of a paired two-factor trajectory

difference depends on rebalancing and on the rule used to couple random noise between equivalent factor representations?

RQ5. **Ranking-tail relevance.** In the trained states, does the exact conditional law of a noisy item-pair margin differ enough from Gaussian approximations to motivate a different ranking method?

These questions concern related but distinct quantities. A difference in score RMS does not automatically imply a difference in ranking relevance; a rank-set change need not lower NDCG; and a smaller transmitted tensor does not establish lower communication to convergence. Each stage therefore uses a measurement appropriate to the claim it can support.

1.3 Completed contributions

The first contribution is a reproducible experimental framework: deterministic data preparation, full-catalog ranking, explicit user-level aggregation and accounting, matched controls, repeated training seeds, and an immutable evidence trail. This framework supports a qualified empirical result: low rank is useful in some strong-privacy comparisons, but it is not a general remedy for private collaborative recommendation.

The second contribution is a controlled recommendation-specific evaluation of effective noise. The study relates factor-space noise to the item product, local user state, clipping, ranking, and communication. The fixed-factor method is a baseline based on established ideas, not a newly invented algorithm. A separately calibrated user-level private popularity method exposes the weakness of several apparently promising comparisons against a deteriorating collaborative reference.

The third contribution is an audited diagnostic finding. Large paired distances in rebalanced two-factor trajectories can depend strongly on coordinate coupling. Alignment and continuation-without-balancing controls substantially narrow the interpretation of the initial local-memory observation. This finding reproduces on a second MovieLens dataset, although that does not establish independent user populations or generalization to a different service.

Finally, the project documents a rejected mechanistic candidate. A correct non-Gaussian conditional margin law produces negligible practical approximation error in the selected real checkpoints. Retaining this result prevents a mathematically interesting construction from being presented as an explanation for utility loss without empirical support. Together, these contributions form a substantial undergraduate research study. Establishing a distinct publishable algorithm or a novel general theorem remains future work.

1.4 Organization

Sections 2 to 4 establish prior work, the threat model, and the evaluation protocol. Sections 5 and 6 describe model variants and experimental controls. Sections 7 to 11 report the completed stages. Section 12 synthesizes the evidence and limitations. The appendices contain derivations, algorithm details, material revisions, and reproduction instructions. The supplement retains the expanded numerical record.

2 Related work and contribution boundaries

2.1 Ranking, federation, and private collaborative learning

BPR learns implicit-feedback rankings by favoring observed items over sampled unobserved items [1]. This project uses its pairwise objective with matrix-factorization scores because the shared item and local user components are easy to identify and audit. It is a diagnostic baseline rather than a claim to represent the strongest contemporary recommender. Federated averaging motivates repeated local optimization and aggregation of client updates [2]. Here clients represent users and hold persistent personalization vectors; the implementation is a centralized simulator of the federated training protocol.

Gaussian perturbation of bounded contributions is a standard DP construction. DP-SGD and user-level federated mechanisms establish the relevant methodological background [3, 4]. Privacy accounting in this work uses numerical privacy-random-variable (PRV) composition, with Rényi DP (RDP) as a separate cross-check [5–7]. The single-release popularity baseline uses analytic Gaussian calibration [8]. The role of client clipping is itself established: its effects on convergence and privacy cannot be inferred from the noise norm alone [9].

Private matrix-factorization research predates this project. Private alternating least squares (ALS) perturbs sufficient statistics for matrix completion [10]; privacy-preserving federated matrix factorization has also been developed explicitly [11]. PrivateRec studies private training and serving in federated news recommendation [12]. These methods differ in objectives, privacy units, trust assumptions, and evaluation tasks. They motivate stronger published comparators, but they were not implemented in the completed experiments. Consequently this report makes no state-of-the-art comparison against them.

2.2 Personalization and public information

Local/global personalization has an existing theoretical foundation under user-level joint DP [13]. Keeping a user-specific vector local is therefore neither a new privacy principle nor evidence that every associated research output is private. The global release, information used for local prediction, and outputs visible to other users must have the appropriate scope. Detailed diagnostic access to local vectors falls outside the training release analyzed here.

Public item features can reduce reliance on private collaborative observations [14]. That work is directly relevant to a possible metadata-plus-private-residual extension. However, E1-FixedB uses an interaction-independent random basis; it contains no genres or semantic movie information. It must not be described as an implementation of a public-content recommender. Work on multi-task privacy under distribution skew also establishes precedents for heterogeneous task weighting and privacy allocation [15]. A future activity-aware method would need to account for that literature and the privacy of any statistics used to choose its weights.

2.3 Low-rank geometry and fixed-factor methods

FFA-LoRA already studies a fixed-factor strategy in privacy-preserving federated low-rank adaptation [16]. FedASK also addresses limitations of fixed-matrix approaches and factor-noise amplification [17]. These are close methodological precedents for E1, even though their application and privacy settings are not identical to whole-user BPR recommendation. The algebraic identity involving AZ_B , Z_AB , and Z_AZ_B cannot supply novelty by itself. Nor can simply fixing one factor.

The geometry of DP perturbations has wider precedents [18]. PRISM explicitly studies gauge-dependent low-rank noise and an intrinsic private mechanism [19]. FLoRG uses Procrustes alignment to address decomposition drift [20]; FedRot-LoRA and FedGSA provide additional neighboring work on rotational alignment and geometry-consistent federated aggregation [21, 22]. The report’s alignment control therefore does not claim a new alignment algorithm. Its empirical role is to test how a chosen paired-noise construction affects personalized recommendation diagnostics.

2.4 Noise propagation and ranking stability

Earlier noise influences later gradients, so a sequence of noisy optimization steps cannot generally be reduced to independent final-weight perturbations. NoiseCurve studies cross-iteration noise using model curvature [23]; DiSK develops noise-reduction ideas for private optimization [24]. These precedents rule out the broad claim that studying how DP noise changes weights is itself an unexplored direction. The more specific question here concerns persistent local user state and the interpretation of paired, factorized trajectories.

Likewise, ranking stability under noisy scores is already an active topic [25]. Products of Gaussian variables and their relationship to differences of noncentral chi-square variables have established distributional theory [26]. E4 applies that theory to a conditional BPR margin and checks practical relevance. It does not derive a new distributional theorem or contradict results proved under explicit sub-Gaussian assumptions.

2.5 How the literature was used

The literature review is a targeted primary-source audit, not a systematic review with an exhaustive search guarantee. The research record distinguishes full method reading from abstract-level screening. The main audit was completed through 3 October 2026, with selected bibliographic checks on 5 October. Formal citation metadata uses 27 retained references. A further screened title with an unresolved date inconsistency remains in the research search log and is excluded from the submission bibliography. Preprints are cited as preprints unless a proceedings source is recorded.

This positioning matters for the interpretation of the whole project. The established components are BPR, federated aggregation, Gaussian DP, fixed-factor training, factor alignment, and product-normal mathematics. The completed project combines them into a controlled recommendation study and a documented diagnostic audit. Whether the narrower measurement finding supports a research paper depends on demonstrating a consequence beyond this particular simulator and comparing it carefully with the closest geometry and alignment work.

3 Problem formulation and privacy model

3.1 Personalized implicit-feedback ranking

Let $\mathcal{I}_u^+$ be user u ’s set of training-positive items. A local vector $p_u \in \mathbb{R}^d$ and shared item row $q_i \in \mathbb{R}^d$ produce score

$$s_{ui} = p_u^T q_i, \quad S = PQ^T. \quad (1)$$

The model has no item biases, content features, attention layers, or temporal features. For a sampled triplet (u, i, j) with $i \in \mathcal{I}_u^+$ and $j \notin \mathcal{I}_u^+$, the implemented BPR loss is

$$\ell(u, i, j) = \log(1 + \exp(s_{uj} - s_{ui})) + \lambda(\|p_u\|^2 + \|q_i\|^2 + \|q_j\|^2). \quad (2)$$

The softplus implementation is numerically stable for extreme margins. The penalty applies to embeddings touched by the sampled loss, not to a separate factor-only penalty or an optimizer's implicit weight decay. In a factorized model, $q_i = A_i B$ and gradients propagate through this effective row.

There are two relevant population objectives. Interaction-weighted centralized training gives more weight to users with more positives. User-weighted training first averages within each user and then across users. Uniform client inclusion and equally weighted client updates are closer to the latter, although local multi-step optimization is not exactly the gradient of a single centralized objective. The B0-UW control makes the weighting difference visible without claiming that it eliminates all optimizer differences.

3.2 Protected information and adjacency

The privacy unit is one user's complete interaction record under add/remove adjacency. For neighboring datasets D, D' differing in one user's contribution, a mechanism $\mathcal{M}$ is (ϵ, δ) -DP when, for each measurable output set $\mathcal{O}$,

$$\Pr[\mathcal{M}(D) \in \mathcal{O}] \leq e^\epsilon \Pr[\mathcal{M}(D') \in \mathcal{O}] + \delta. \quad (3)$$

The catalog, eligible population size used for normalization, configurations, and random public basis are treated as fixed public inputs. Eligibility filtering and catalog construction in the research data pipeline are not themselves privatized. Thus the reported mechanism is conditional on these inputs; it is not an end-to-end private data-ingestion and model-selection system.

User vectors persist locally and are never included in the shared update. The trust model assumes a server that sees only a noised aggregate, as would be supported by suitable secure aggregation and noise generation. The simulator does not implement a cryptographic secure-aggregation protocol. Raw client updates or local checkpoints visible to an untrusted server would not be protected merely by calling the training procedure federated.

3.3 Private aggregation mechanism

Write θ_t for the shared parameter vector: vectorized Q , (A, B) , or A alone. At round t , each client is independently sampled with probability q . A selected client starts from θ_t , performs local updates using its own data and local p_u , and returns shared difference $\Delta_{u,t}$. Clip the whole shared contribution once:

$$\bar{\Delta}_{u,t} = \Delta_{u,t} \min\left(1, \frac{C}{\|\Delta_{u,t}\|_2}\right), \quad \theta_{t+1} = \theta_t + \frac{\eta_s}{qN} \left(\sum_{u \in \mathcal{U}_t} \bar{\Delta}_{u,t} + \xi_t \right), \quad \xi_t \sim \mathcal{N}(0, \sigma^2 C^2 I). \quad (4)$$

For Two, the norm is over the concatenated factor update. Clipping each factor independently at C would define a different sensitivity bound. Noise is added to every shared coordinate, including item rows untouched by sampled users. The denominator is the fixed expectation qN , not the realized number of participants. Empty private rounds still release a noised update and count toward composition. Local states are retained only by their respective users.

Table 1: Final iterative privacy calibration at $q = 0.1$, $T = 50$, $\delta = 10^{-5}$.

ϵ target	σ	PRV bound	RDP cross-check
8	0.805072021	7.967513	9.119818
4	1.151928711	3.963885	4.497283
2	1.760449219	1.993278	2.220645
1	2.977490234	0.990062	1.087196

PRV is primary. RDP is a separate bound on the same mechanism. Source: `privacy_accounting.csv`, final 50-round rows.

At the ML-100K population size, $qN = 94.2$. At ML-1M, $qN = 603.5$. The primary accountant depends on q, T, σ, δ , so transferring these parameters preserves the accountant bound while changing the post-normalization noise SD. Changing dimension alone does not reduce ϵ . Neither does changing C while scaling Gaussian noise by the same multiplier. Utility can change substantially in both cases.

All final private collaborative experiments use $T = 50$, $q = 0.1$, $C = 1$, and $\delta = 10^{-5}$. Table 1 reports the exact retained calibration. PRV is the declared primary bound. The RDP column is a distinct upper bound for the same mechanism, not an extra privacy charge and not a replacement selected after inspecting utility.

3.4 Release scope and local personalization

The guarantee describes one fixed-hyperparameter training mechanism. Publishing multiple runs requires composition; selecting a configuration from private validation results is not free; reporting user-level diagnostics is not covered by the final aggregate’s guarantee. The research process uses nonprivate validation, evaluation, and access to internal states. Reproducible random seeds are useful for auditing but do not constitute production-grade secure randomness.

Local personalization can be interpreted through the established joint-DP framework when a user’s output depends on that user’s own records and an appropriately private shared transcript. This does not authorize publishing the entire local matrix P . E2 and E3 use oracle reset and alignment operations on private states for diagnosis; E4 conditions on private checkpoints. They are research measurements, not releases justified by the starting checkpoint’s privacy label. These distinctions prevent the formal accountant from being mistaken for an end-to-end privacy claim about the report itself.

4 Datasets and evaluation protocol

4.1 MovieLens-100K and implicit conversion

The initial benchmark uses the official MovieLens-100K release [27]. Ratings of four or five are treated as positive implicit feedback. Ratings below four are not positive training events, but they remain part of the observed history used for candidate filtering. No rating-regression target is optimized. Users require at least three positives so that one training positive and two held-out positives can remain. One user with no positives is removed. Items are not removed for having few or no training positives.

Each user’s positives are ordered by timestamp. The last positive becomes the test target, the penultimate positive the validation target, and the rest training observations. Equal timestamps

Table 2: Dataset sizes under the frozen chronological implicit-feedback protocol.

Quantity	MovieLens-100K	MovieLens-1M
Eligible users	942	6,035
Catalog items appearing in ratings	1,682	3,706
Training positives	53,491	563,206
Validation targets	942	6,035
Test targets	942	6,035
Test targets scored in this project	942	0

The ML-100K raw release contains 943 users and 100,000 ratings; 55,375 ratings are positive. ML-1M has 575,281 positives before eligible-user filtering; five users are removed. The ML-1M catalog is the set represented in the ratings file, not every row of a metadata catalog.

Source: Frozen dataset report and E3 dataset provenance JSON.

use a deterministic SHA-256 tie breaker based on user ID, item ID, and split seed 2026. This is a per-user chronology, not a global temporal split across all users. Ties cannot recover the actual order of simultaneous events. On ML-100K, 316 users have a validation timestamp equal to their final training timestamp, and 305 have equal validation and test timestamps; the fixed convention is therefore consequential and documented.

The retained ML-100K users have a mean of 56.7845 and median of 37.5 training positives, with a range of 1–376. Activity groups are defined solely from these counts: low activity has at most 22 positives (319 users), medium activity 23–61 (310 users), and high activity at least 62 (313 users). Six validation and fourteen test targets are cold items with no training-positive observation across the population. They remain in the principal evaluation.

4.2 Negative sampling and candidate filtering

For training, a negative is sampled uniformly from items outside that user’s training-positive set. The sampler does not consult validation or test labels. Consequently a future positive can occasionally be sampled as unobserved; filtering it out would use held-out information. Low-rated items can also be sampled as ordinary nonpositives. This convention is distinct from the evaluation filter.

For each held-out event, evaluation ranks the complete catalog after removing all items the user rated before that event, including low-rated items. The target is retained. Later ratings never filter an earlier event’s candidate set. Validation and test therefore have different historical filters, and the test history includes the validation event. On ML-100K the mean candidate counts are 1,581.5658 and 1,579.0658 respectively. There is no sampled-negative evaluation that could make ranking artificially easy or alter comparability between methods.

Metric ties are resolved pessimistically for the relevant target. Diagnostic top-10 sets in E2–E4 instead use a deterministic item-index tie rule and exclude training positives only. Those label-free diagnostic candidate sets are not interchangeable with the historical held-out evaluator.

4.3 Ranking metrics

With one relevant held-out item per user and its one-based rank R_u ,

$$\begin{aligned} \text{HR@10}_u &= \mathbb{I}\{R_u \leq 10\}, \\ \text{NDCG@10}_u &= \frac{\mathbb{I}\{R_u \leq 10\}}{\log_2(R_u + 1)}, \quad \text{MRR@10}_u = \frac{\mathbb{I}\{R_u \leq 10\}}{R_u}. \end{aligned} \quad (5)$$

Metrics are averaged equally over eligible users. Recall@10 equals HR@10 in this one-target design, so it does not add an independent outcome. NDCG@10 is the declared primary measure. HR@10 and MRR@10 are retained to expose changes in hit rate and rank position; they are secondary, without an additional family of confirmatory significance claims.

4.4 MovieLens-1M replication

E3 obtains MovieLens-1M from the official GroupLens distribution and applies the unchanged conversion and splitting functions. The archive passes its published MD5 and ZIP integrity checks, and raw and processed SHA-256 hashes are recorded. Table 2 gives the resulting sizes. E3 observes a fixed sample of 128 users for trajectory diagnostics while allowing every eligible user to participate in training. Full-population validation is scored for the 30 trained models; the test set is not scored.

Both datasets originate from MovieLens. Their user or catalog overlap has not been audited. Replication on the larger dataset is evidence that the diagnostic is not confined to one saved small-data training run; it is not proof of independent populations or cross-domain generalization.

4.5 Holdout exposure and dataset limitations

The ML-100K test results have been used repeatedly across historical stages. An earlier regression-test incident also performed nine two-round real-data B2 test evaluations. Those accidental outputs were not inspected for selection, but their existence means the benchmark cannot be called a pristine holdout. Later stages freeze their own choices before scoring, preserving a meaningful local separation of selection and evaluation without erasing this history.

MovieLens is a public research dataset rather than a deployment privacy population. Positive-threshold conversion, per-user holdout, missing-not-at-random exposure, and very few cold targets all constrain interpretation. The report does not infer a reliable cold-start improvement from fourteen examples, nor does it treat offline rank utility as user satisfaction in a live service.

5 Models and implementation

5.1 B0 and B0-UW: centralized controls

B0 learns full 64-dimensional user and item embeddings centrally using BPR, with Gaussian initialization of SD 0.01. Its final retained configuration is Adam learning rate 10^{-3} , explicit L2 coefficient 10^{-2} , batch size 1,024, at most 1,000 epochs, and patience of 100 validation checks. B0-UW uses the same modeling and optimization settings but samples a user uniformly and then a positive uniformly within that user. B0 samples interactions uniformly. The matched three-seed experiment separates the change in user weighting from the change to federated optimization more clearly than B0-B1 alone.

Earlier B0 documents retain superseded learning rates and stopping rules. The report uses the later common-stopping-protocol results as the final reference. Historical search tables remain part of the audit trail rather than being blended into a new average.

5.2 B1: federated full matrix without DP

In each B1 round, users are independently included with $q = 0.1$. A selected user starts from the current shared Q and its persistent local vector, then performs two local epochs on its own positives. The local SGD learning rate is 5, regularization is 10^{-5} , and the server learning rate is 1. The large numerical difference from B0’s Adam rate reflects a different optimizer and the normalization of each local loss; it is not a directly comparable step-size scale. Selected users’ item differences are averaged with equal client weight. B1 divides by realized participation and skips empty rounds because it has no private-release requirement.

Validation occurs every ten rounds. Patience is 100 validation checks, with an 8,000-round maximum, and the best validation checkpoint is restored for evaluation. B1 is the converged nonprivate federated reference. Local user vectors are not transmitted, but neither federation nor local storage supplies an additional formal guarantee in this nonprivate baseline.

5.3 B2: private full matrix

B2 applies [equation \(4\)](#) to the complete shared item update while retaining B1’s local training settings. Its final private horizon is 50 rounds with $C = 1$ and server rate 1. It always evaluates the final round, without privacy-dependent early stopping. The fixed denominator, full-coordinate noise, and empty-round accounting distinguish it from a simple “add noise to B1” implementation.

The earlier B2 protocol used 1,000 rounds, $C = 1.5$, and server rate 2. Validation studies motivated the final 50-round protocol. Three settings changed together, so the improvement cannot be attributed uniquely to shorter training. The retained no-DP control at the 50-round horizon is necessary: the gap from a converged B1 includes a training-budget effect even when no noise is present.

5.4 B3 and B4: jointly trained item factors

B3 represents $Q = AB$ at ranks 4, 8, 16, 32 and trains both factors without DP. User vectors retain dimension 64. Regularization is applied to p_u and the effective item rows $A_i B$, with gradients through both factors. Adding independent L2 penalties on A and B would change the objective and the role of scale, so that is not done.

QR decompositions and a core SVD rebalance the factors while preserving their product up to floating-point tolerance. This avoids an extreme reciprocal scaling of A and B , but it does not remove a perturbation already present in AB . It also introduces a changing orthogonal coordinate convention, which becomes important in the later trajectory probes. The final B3 local rates are 0.625, 1.25, 1.25, 0.625 in increasing rank order. B3 follows B1’s convergence rule with one disclosed exception: rank 32, seed 123 uses a predeclared 16,000-round diagnostic ceiling and stops at 9,350, with best checkpoint 8,350. All other final runs stop below the original ceiling.

B4 applies a single joint clip to $(\Delta A, \Delta B)$, then independent Gaussian coordinates across that concatenated vector. It uses the same q, T, C, σ, δ as final B2. Its final rank-4 local/server

rates are 2.5/1; the other ranks use 5/0.5. Historical B4 received more validation tuning than B2, including a bounded fallback after unstable candidate trajectories. The B2–B4 comparison is therefore a comparison of retained methods, not a clean intervention on coordinate count alone.

5.5 E1: a separately controlled comparison

E1 preserves B0–B4 and creates three explicit families: Full, Two, and FixedB. Full repeats the final full-matrix mechanism; Two trains both factors; FixedB holds a public row-orthonormal B fixed and privately trains only A . The basis is seeded independently of interactions (public seed 314159). At each rank, Two and FixedB begin with the same effective Q . This removes an avoidable initialization discrepancy between the two factorized methods.

Each of nine units (Full and two methods at four ranks) receives four local/server learning-rate candidates: $(1.25, 1)$, $(2.5, 1)$, $(5, 0.5)$, and $(5, 1)$. Selection uses three seeds at reference privacy target four, subject to finite behavior across six levels: no clipping/no noise, clipping only, and the four private targets. The selected local rate is 5 throughout. The server rate is 1 for Full and FixedB, 0.5 for Two ranks 4/8/16, and 1 for Two rank 32. FixedB’s local rate reaches the upper grid boundary, a limitation on claims about its optimal attainable utility.

The fixed public basis is useful analytically. Let $v_u = Bp_u$. Then $s_{ui} = A_i^T v_u$, and every $v_u \in \mathbb{R}^r$ can be realized by $p_u = B^T v_u$ when $BB^T = I_r$. Thus FixedB retains the same hard score-rank bound as Two at rank r ; its failure cannot be attributed simply to a smaller maximum score rank. Jointly learning B changes the latent metric, gradient dynamics, clipping, and aggregation behavior. It does not increase the maximum rank beyond r . Appendix A gives the equivalence precisely.

5.6 User-level private popularity

For each user define a binary indicator $h_u \in \{0, 1\}^M$ for distinct training-positive items, and bound its contribution by

$$x_u = h_u \min \left(1, \frac{\sqrt{20}}{\|h_u\|_2} \right), \quad z = \sum_u x_u + \mathcal{N}(0, 20\sigma_{\text{pop}}^2 I). \quad (6)$$

The sum has add/remove user sensitivity $\sqrt{20}$. A single full-population release is calibrated analytically at the same target (ϵ, δ) as the iterative models, using its own multiplier. Items are ranked by z_i , with the same user-specific held-out candidate filter. This is a useful nonpersonalized collaborative reference because it spends its budget on one bounded statistic instead of repeated iterative updates.

The no-noise bounded-popularity score is not the same as raw item counts: high-activity users are downweighted by the norm bound. The historical raw popularity, bounded no-noise popularity, and private popularity must therefore be named separately. A local metadata-only model and a public-features-plus- private-residual model were proposed as future directions but were not trained.

Table 3: Experimental stages and their evidence units.

Stage	Question	Retained evidence
Phase 0	Can the ranking protocol be reproduced?	Data fingerprints, evaluator checks, random and raw-popularity references.
B0–B4	How do weighting, federation, privacy, and rank interact?	Three-seed weighting control; five-seed final federated comparisons.
E1	Does a fixed public factor help under equal tuning?	648 search jobs; 270 final collaborative states; 21 popularity arrays.
E2/E2b	Is local-state disturbance persistent and coupling-sensitive?	120 pulse/reset probes and 40 aligned follow-up probes from 30 checkpoints.
E3	Does the diagnostic replicate and depend on balancing?	30 ML-1M training runs; 200 controlled probes across two datasets.
E4	Do non-Gaussian margins matter in trained states?	50 states; 3,200 prescribed item pairs; six synthetic cases.

A probe is not an independently trained model. Some E1 final states reuse eligible search runs, and E2–E4 deliberately reuse frozen checkpoints.

5.7 Communication accounting

The simulator counts dense float32 uploads and downloads for participating clients, excluding local P . At a client-round, the modeled payload is

$$b_{\text{Full}} = 8Md, \quad b_{\text{Two}} = 8r(M + d), \quad b_{\text{FixedB}} = 8Mr \quad \text{bytes.} \quad (7)$$

The factor eight is four bytes per scalar in each direction. FixedB can reconstruct its public basis from a seed; explicit provision would add a one-time $4rd$ bytes per client. Network framing, compression, cryptographic overhead, failures, and real device latency are not measured. Total modeled traffic accumulates the payload over realized selected clients and rounds. It is distinct from payload size and from utility under a fixed total budget.

6 Experimental design and statistical analysis

6.1 Sequential controls and preserved stages

The historical ladder changes one broad design component at a time, while explicit controls expose remaining differences. B0–UW addresses weighting; 50-round no-noise controls address undertraining; E1 clipping-only controls separate clipping from additional perturbation; and the private popularity reference tests whether iterative personalization is competitive at all. No single comparison isolates every causal mechanism. In particular, parameterization changes optimization and sensitivity geometry together.

E1’s grid has $9 \times 4 \times 3 \times 6 = 648$ search jobs, including eleven retained failures. The selected units, six levels, and five seeds yield $9 \times 6 \times 5 = 270$ finite final states. Reuse of eligible search cells gives 756 unique training records rather than 918. Twenty noisy popularity arrays and one deterministic bounded no-noise array bring one-time extension test scoring to 291 models. Selections and validation-chosen ranks were frozen at 15:37:41 UTC on 2 October 2026, before the first extension test scoring. This freeze is a stage-specific safeguard, not evidence that the dataset had never been examined earlier.

6.2 Seeds and paired uncertainty

The centralized weighting comparison uses seeds 42, 123, and 2026. Final B1–B4 and E1 use those plus 7 and 99. Utility tables give the arithmetic mean and sample SD across the retained seeds. To compare two methods on NDCG@10, first average the metric over seeds separately for each user, then form the paired user difference. A bootstrap samples users with replacement from these paired differences 100,000 times with bootstrap seed 2026.

The primary B3 family contains four comparisons against B1; B4 contains sixteen rank-by-privacy comparisons against B2; E1 contains sixteen matched-rank FixedB-minus-Two comparisons. Bonferroni-adjusted interval levels are 98.75% for B3 and 99.6875% for each sixteen-comparison family. These are family-wise interval constructions conditional on the saved training runs and population represented by the benchmark. They do not jointly quantify uncertainty over new datasets and independently sampled training procedures.

An adjusted interval entirely above zero supports a positive comparison within its family. An interval containing zero is recorded as “includes zero,” not as proof of equivalence. Ordinary 95% intervals for secondary comparisons are retained in the supplement but do not inherit the primary multiplicity interpretation. Figure error bars are seed SDs, and their overlap is not the criterion used for primary decisions.

6.3 Rank selection, groups, and personalization diagnostics

For presentation of a single low-rank curve, each method’s rank is selected from validation at the relevant noise level. That curve is different from the matched-rank primary comparison. A test-best rank never replaces a validation-selected rank because it happens to look better in the final table. The historical B4 and E1 selections differ and are retained separately.

Activity and cold-target tables are descriptive. Their groups are not independent discoveries, and the fourteen cold targets are too few for a strong cold-item claim. E1 also compares validation predictions using each user’s own vector, permuted vectors, and a mean vector. These 810 retained rows diagnose the contribution of local personalization. They are oracle analyses of stored states, not alternative serving procedures with analyzed privacy guarantees.

6.4 Diagnostic replication and advancement criteria

E2 averages two pulse replicates within a checkpoint seed before summarizing across five seeds. E2b follows the same rule. Their means, medians, ranges, and SDs are descriptive; correlated users, intervention branches, and draws are not treated as independent training runs. E3 keeps transferred training settings fixed and uses a preselected observer population. E4 declares its checkpoint set, pair types, and practical-error gate before examining outputs. The gate requires absolute matched-Gaussian probability error greater than 0.01 on at least 5% of primary pairs in both datasets. It is a research screening criterion, not a significance threshold.

Protocols, source hashes, checkpoint identities, failure inventories, and observations are retained. This separation of initial evidence, adaptive follow-ups, and final decisions makes the chronological research process auditable without pretending every later question was specified at the outset.

Table 4: Matched centralized and federated controls (three seeds).

Model	NDCG@10	Seed SD	HR@10	MRR@10
B0	0.085972	0.000456	0.171975	0.060051
B0-UW	0.091745	0.001866	0.181882	0.064536
B1	0.087539	0.002074	0.164190	0.064527

Seeds 42/123/2026. This B1 mean differs from the five-seed B1 mean used in later comparisons. Source: centralized_federated_3seeds.csv.

Table 5: Converged nonprivate federated utility (five seeds).

Model	NDCG@10 $\pm$ seed SD	Adjusted interval vs. B1
B1	0.08901 $\pm$ 0.00249	Reference
B3 r4	0.05947 $\pm$ 0.00381	[-0.041779, -0.017893]
B3 r8	0.07101 $\pm$ 0.00596	[-0.027511, -0.008949]
B3 r16	0.07633 $\pm$ 0.00722	[-0.020950, -0.004560]
B3 r32	0.06807 $\pm$ 0.01354	[-0.031093, -0.011160]

Intervals are 98.75% paired-user bootstrap intervals for the four-comparison B3 family. Source: final_5seed_means.csv and paired_contrasts.csv.

7 Results of the historical baselines

7.1 Sanity references and centralized learning

Phase 0 established random-ranking and raw-popularity references under the same full-catalog evaluator later used for learned models. The historical nonprivate raw-popularity test NDCG@10 is 0.044292; random ranking is much lower, approximately 0.0038. These references help distinguish personalized learning from a pipeline that merely returns popular items or fails to learn. The full exact sanity metrics are retained in the curated results archive.

Table 4 gives the final three-seed comparison. Centralized B0 reaches 0.085972, B0-UW 0.091745, and B1 0.087539 test NDCG@10. The paired B0-UW-minus-B0 difference is 0.005773, with ordinary 95% interval $[-0.000164, 0.011910]$. B1-minus-B0-UW is -0.004206 , with interval $[-0.010903, 0.002288]$. The total B1-minus-B0 difference is 0.001567, with interval $[-0.005353, 0.008668]$. All three overall intervals contain zero.

The point estimates suggest that user weighting partly offsets the cost of federated optimization, but this study does not resolve an overall superiority claim for either change. In particular, B1’s numerical proximity to B0 should not be used to assert that federation has no effect: the intervention changes weighting, optimizer, and training organization together. B0-UW improves the interpretability of that comparison while preserving the uncertainty.

Historical personalization checks replace a user’s own embedding with another user’s embedding and substantially reduce utility. The saved B1 seed-42 check falls from approximately 0.0890 to 0.0202 NDCG@10. This is a sanity check that the model uses user-specific information, not a five-seed estimate of a new serving method. E1 later broadens this diagnostic on validation states.

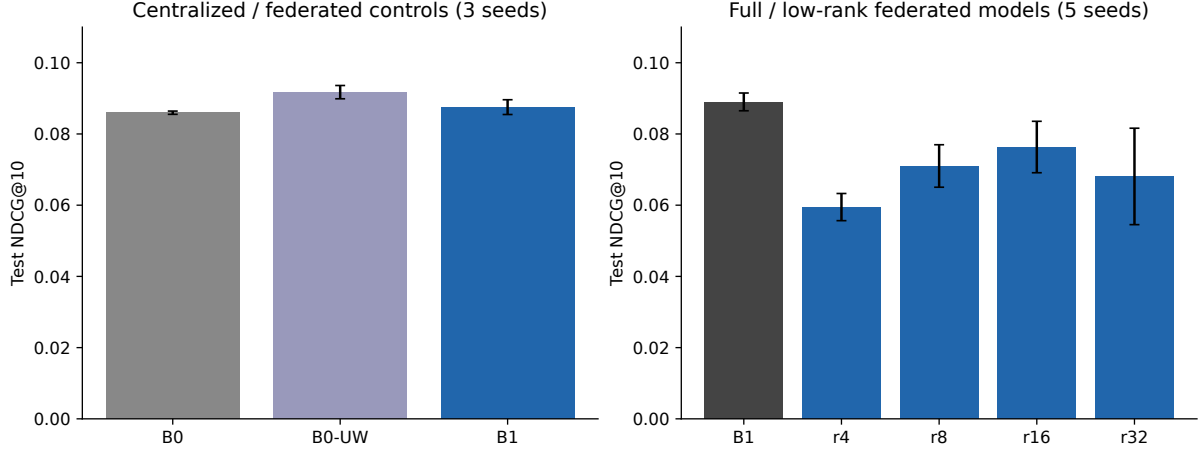


Figure 1: Left: matched three-seed centralized and federated test NDCG@10. Right: five-seed converged B1 and B3. Whiskers show seed SD. The panels have different seed sets. Overall weighting-decomposition intervals include zero; all four adjusted B3-versus-B1 intervals are negative. Sources: retained centralized control and final five-seed tables.

Table 6: Historical private and matched no-DP test NDCG@10 (five-seed means).

Target / control	B2 Full	B4 r4	B4 r8	B4 r16	B4 r32
No DP	0.057889	0.040834	0.048083	0.049143	0.050931
8	0.052271	0.038544	0.045385	0.043628	0.044022
4	0.042169	0.037846	0.042653	0.038274	0.044489
2	0.025428	0.029079	0.034485	0.033993	0.036504
1	0.011461	0.015976	0.018162	0.016685	0.015390

All rows use 50 rounds. Historical tuning opportunities differ. Seed SDs and secondary metrics appear in the supplement. Source: final_5seed_means.csv.

7.2 Nonprivate low rank

The five-seed converged B1 mean is 0.089008, which differs from the three-seed B1 entry because the seed set differs. The B3 means at ranks 4, 8, 16, and 32 are 0.059468, 0.071010, 0.076333, and 0.068074. All four adjusted intervals favor B1 (table 5). Low rank therefore incurs a utility cost before DP is introduced under these retained settings.

The rank trend is not monotonic. Rank 16 has the largest B3 mean; rank 32 has larger seed variation and documented convergence/stability complications. This is evidence about the trained models, not a theorem that rank 32 has less representational capacity. Optimization, selected rates, and stopping behavior contribute to the outcomes. The expanded rank-32 diagnostic ceiling is disclosed rather than hidden inside an apparently common training budget.

7.3 Private full matrix and low rank

The full-matrix B2 mean falls from 0.057889 in its matched 50-round no-DP control to 0.052271, 0.042169, 0.025428, and 0.011461 at privacy targets 8, 4, 2, and 1. Its no-DP value is already below converged B1, so the entire B1–B2 gap cannot be assigned to privacy noise. B4’s no-DP controls also start below B2’s no-DP value.

The sixteen adjusted B4-minus-B2 comparisons give a conditional crossover: all four ranks

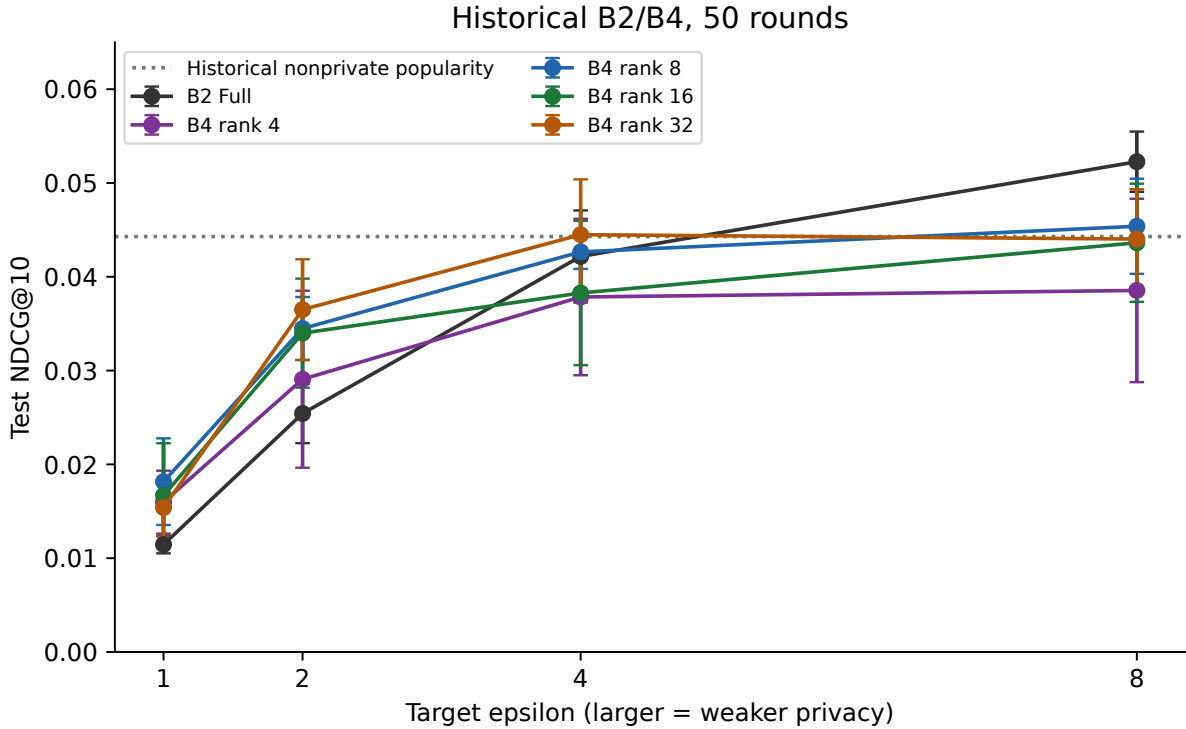


Figure 2: Five-seed test NDCG@10 with seed-SD whiskers for B2 and all B4 ranks at 50 rounds. Larger epsilon means weaker privacy. The dotted popularity line is nonprivate raw popularity. Historical tuning budgets differ. Four of sixteen adjusted B4-minus-B2 comparisons are positive; line segments only connect discrete experimental settings.

lose at target eight; no adjusted difference is detected at target four; ranks 8, 16, and 32 win at target two; and rank 8 wins at target one. These are the four positive comparisons. Several additional ordinary 95% intervals look favorable but do not survive the declared adjustment. The validation-selected rank at target one is 16, not the test-best rank 8; the former’s adjusted interval includes zero. Selecting the latter after examining test scores would change the interpretation of the experiment.

At target two, for example, B4 rank 32 reaches 0.036504 against B2’s 0.025428. Yet both are below the historical raw-popularity point reference. The low-rank model’s smaller loss from its own weaker no-DP control does not by itself establish better recommendation quality. This observation motivates the user-level private popularity comparator introduced in E1.

7.4 Communication and the original hypothesis

The rank-8 two-factor payload is 111,744 bytes per participating client-round, compared with 861,184 bytes for Full. At a fixed 50-round horizon, this approximately 7.7-fold payload reduction also produces a corresponding reduction in modeled total tensor volume. B2 averages 4.026 GB and B4 rank 8 about 0.522 GB. These are meaningful savings in the simulator’s accounting.

Converged training has a different comparison. B1 averages 93.911 GB to its best checkpoint; B3 rank 32 averages 180.329 GB despite a smaller per-round payload. The latter requires many more rounds. Rank 4 and rank 8 do reduce total volume, but their utility is lower. The full communication table is in appendix B and the supplement. No experiment in this project optimizes utility under an identical end-to-end communication cap.

The historical results therefore partially support the initial hypothesis: low rank can offer a better private point on the utility/payload trade-off, especially when the full model is heavily degraded. They reject its strongest form, namely that fewer shared noisy parameters reliably improve utility. Unequal historical tuning and multiple changes in optimization prevent a dimension-only causal interpretation. E1 was designed to examine the proposed noise explanation more directly.

8 E1: effective noise and fixed-factor training

8.1 Why the effective matrix matters

Condition on the signal-updated factors and the just-updated user matrix. Let Z_A and Z_B have independent, identically distributed Gaussian entries with SD $\tau = \eta_s \sigma C / (qN)$. The immediate effective-matrix shock is

$$D = (A + Z_A)(B + Z_B) - AB = AZ_B + Z_AB + Z_AZ_B. \quad (8)$$

The three terms have zero mean and zero expected pairwise Frobenius inner products, but they do not all need to be independent. Their expected energies give

$$\mathbb{E} \|D\|_F^2 = \tau^2 (M \|B\|_F^2 + d \|A\|_F^2) + Mdr\tau^4, \quad (9)$$

$$\mathbb{E} \|PD^T\|_F^2 = \tau^2 (M \|PB^T\|_F^2 + \|A\|_F^2 \|P\|_F^2) + Mr\tau^4 \|P\|_F^2. \quad (10)$$

Thus coordinate dimension is only one determinant of prediction disturbance. A method with modest effective item noise can still have a large score shock if its local user vectors grow. Conversely, multiplying all of a user's scores by a positive constant changes their scale without changing their ranking. Norms, score RMS, margins, and ranking outcomes must be interpreted together. These conditional calculations do not describe the full distribution after many rounds of state-dependent learning.

FixedB removes the two terms involving a noisy B . With $BB^T = I_r$, $D = Z_AB$ and $\|D\|_F = \|Z_A\|_F$. This is a clean linear diagnostic baseline. It also changes clipping geometry and aggregation behavior, so better conditional noise moments do not guarantee better trained utility. Eight synthetic Monte Carlo checks corroborate the implemented moment calculations; derivations and assumptions are in [appendix A](#).

8.2 Controlled utility comparison

The primary family compares FixedB with Two at each matched rank and privacy level. None of the sixteen adjusted comparisons favors FixedB. Eight favor Two and eight include zero. At

Table 7: E1 test utility with ranks selected on validation at each privacy level.

ϵ	Full	Two	FixedB	DP popularity
8	0.052271	0.051158 (r32)	0.048978 (r32)	0.049548
4	0.042169	0.045807 (r32)	0.038959 (r32)	0.049428
2	0.025428	0.040841 (r8)	0.025286 (r16)	0.049507
1	0.011461	0.025136 (r4)	0.011958 (r16)	0.048389

Five-seed NDCG@10 means. The primary E1 family compares matched ranks, not these separately selected ranks. Source: `test_means.csv` and `rank_selection.csv`.

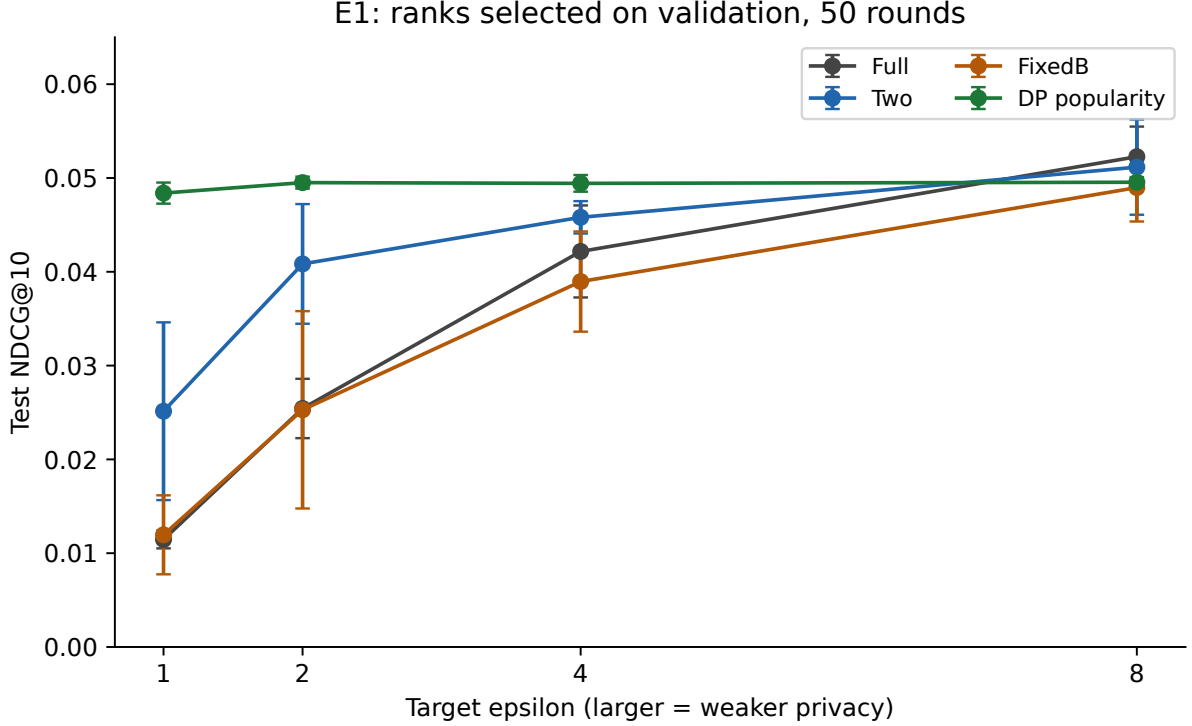


Figure 3: Full, Two, FixedB, and private popularity test NDCG@10 with five-seed SD. Ranks for Two and FixedB are selected separately on validation at each level. The primary tests instead compare matched ranks. Collaboration uses 50 rounds and popularity one separately calibrated bounded release. Every collaborative point mean at target epsilon at most four is below private popularity; this is not a family-wide significance claim against popularity.

target two, FixedB is worse at all four ranks. The rank-8 difference is -0.018662 , with adjusted interval $[-0.027695, -0.010368]$. This is stronger evidence against the proposed standalone fixed-factor improvement than an informal comparison of two selected example runs.

The validation-selected presentation in table 7 leads to the same practical conclusion, while answering a different question. Two chooses ranks 32/32/8/4 at privacy targets 8/4/2/1; FixedB chooses 32/32/16/16. Two reaches 0.040841 at target two and 0.025136 at target one, whereas FixedB reaches 0.025286 and 0.011958. These selected-rank curves should not be described as the matched-rank primary statistical family.

FixedB can perform well in no-noise controls, and it often has smaller immediate effective noise. Its poorer private utility therefore does not show that the implementation lacks all ranking capacity. It shows that removing one noise source, with these transferred clipping and search choices, does not overcome the remaining optimization and learning limitations. The grid boundary, parameterization-dependent clipping, and different selected server rates remain qualifications on any broader statement about optimal fixed-factor performance.

8.3 The private popularity baseline changes the reference point

The bounded no-noise popularity model obtains 0.049885 test NDCG@10. Its private mean at target one is 0.048389 ± 0.001125 , retaining about 97% of that bounded reference. The corresponding means at targets 8, 4, and 2 are 0.049548, 0.049428, and 0.049507. All E1 collaborative point means at targets at most four lie below the corresponding private popularity mean. This

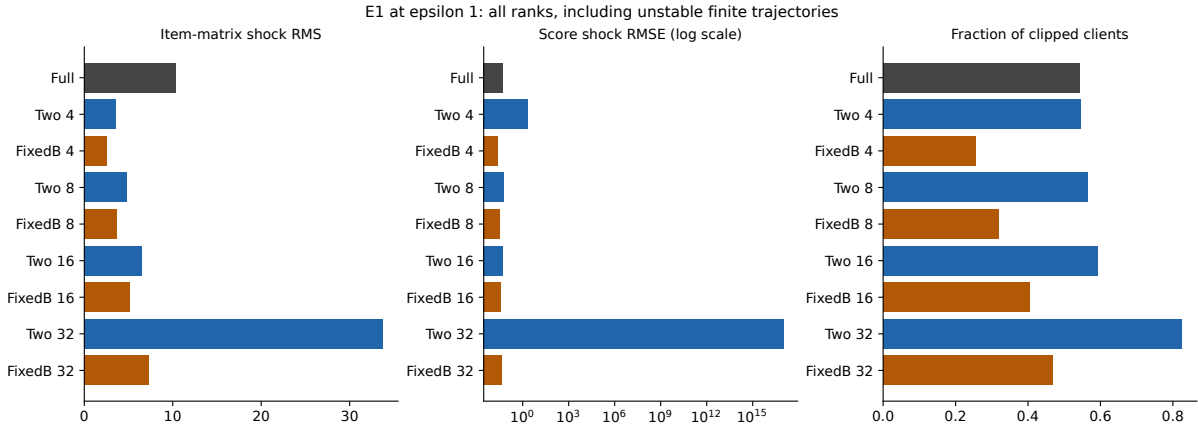


Figure 4: E1 diagnostics at target epsilon one. Left: mean over seeds of each run’s root-mean-squared Frobenius norm of the immediate effective item shock. Center: mean score-shock RMSE, on a logarithmic axis retaining the large finite Two-r32 outlier. Right: selected-client clipping fraction. These aggregates are diagnostics, not direct causal measures of NDCG loss. Source: `geometry_means.csv`; per-run evidence remains available.

is a descriptive statement covering the retained settings, not a multiplicity-adjusted superiority theorem.

The comparison is useful even though the mechanisms allocate their budgets differently. Both protect an entire user’s bounded contribution at the reported target, but popularity uses one full-population release and does not attempt personalized collaborative representation learning. Its strength suggests that the useful private signal in these settings is not exhausted by the iterative methods’ current optimization. It does not prove that personalization is fundamentally impossible under the same privacy target.

8.4 Measured geometry, clipping, and numerical behavior

At target one, the Full effective-noise norm is approximately 10.369. FixedB ranks 4/8/16/32 have norms 2.592, 3.667, 5.186, and 7.333; the corresponding Two values are approximately 3.602, 4.817, 6.542, and 33.788. These are averages of per-run RMS Frobenius norms over rounds, not per-entry noise standard deviations. Different selected server rates also affect the comparison.

The expected bilinear term accounts for only about 0.2–0.3% of total effective-matrix noise energy in the selected Two settings at target one. Its presence is algebraically important, but its measured energy does not support calling it the dominant cause of the observed utility failures. First-order terms, factor scales, clipping, and local learning remain relevant.

Clipping at the same coordinate threshold affects the methods differently. At target one, approximately 54.1% of selected Full updates are clipped; FixedB fractions range from 25.6% to 46.9%; Two fractions range from 54.6% to 82.5%. These fractions accompany different effective signal steps, not just different noise counts. Equal C across parameterizations does not mean equal effective- Q sensitivity or equal retained signal.

Some Two trajectories have very large but finite local user norms. The mean score-shock RMSE for Two-r32 at target one is approximately 9.846×10^{16} , dominated by extreme state scale. An original float32 norm summary overflowed even though stored vectors were finite; analysis was corrected by reading those vectors in float64. Original logs, training, and held-out ranks were preserved. The large values are reported rather than excluded, but a mean dominated

by such an outlier is not a typical-user effect. Ranking diagnostics and seed-level distributions are necessary companions to the RMS summary.

8.5 Activity groups and personalization ablations

At target two, validation-selected Two-r8 obtains test NDCG@10 of 0.048722, 0.027027, and 0.046489 in the low-, medium-, and high-activity groups. Private popularity gives 0.055746, 0.036872, and 0.055662. Thus the aggregate gap in this comparison is not explained solely by one activity group. These are descriptive group means, without an adjusted subgroup test. The cold-target group is too small for a reliable comparison: Two-r8 and popularity both score zero there at target two, while Full has a small nonzero value from very few top-ten outcomes.

Local-vector ablations expose a further distinction between storing a user vector and obtaining useful personalization from it. In the Full no-DP control, validation NDCG is 0.053136 using the user’s own vector, 0.042725 with permuted vectors, and 0.047421 with the mean vector. At target one the corresponding values are 0.008883, 0.011147, and 0.030485. FixedB-r8 likewise has little own-versus-permuted difference at target one (0.008580 versus 0.008481), while its mean-vector diagnostic is 0.018912. These contrasts show that local parameter persistence alone does not guarantee a personalization benefit under noise.

The mean-vector calculation uses the collection of private local states and is evaluated retrospectively. It is not a free public statistic, a deployable private method, or a new selected baseline. There is no confirmatory ablation significance claim. Its role is to motivate further investigation of how local user learning interacts with the noisy item representation, which E2 studies through explicit interventions.

8.6 Recommendation-specific implications

The item factor is strongly rectangular: A has $1682r$ entries and B only $64r$. Fixing B therefore saves $64/(1682 + 64) = 3.67\%$ of repeatedly shared coordinates relative to Two at the same rank, not half the payload. At rank 8 the FixedB payload is 107,648 bytes versus Two’s 111,744. Most savings relative to Full come from low rank, which both methods already use.

The activity, cold-target, and local-vector ablation results are retained in the supplement. They provide descriptive information about heterogeneous personalization, but they do not establish a subgroup-specific improvement. In particular, a random public basis supplies no semantic information for an unobserved item. A metadata-only personalizer and a small private residual remain distinct future comparisons. E1’s substantive outcome is the failure of a plausible fixed-factor improvement under controlled testing, together with a clearer set of measurements for investigating why.

9 E2/E2b: local-state memory and noise coupling

9.1 Question and intervention design

Immediate noise moments hold the local user matrix fixed. During training, however, users adapt to the shared item state. A shared perturbation can therefore change local vectors, which later influence the shared model again. E2 asks whether that local-state pathway has measurable prediction consequences after the shared state is restored. It is a mechanism probe, not a proposed private denoising algorithm.

The study starts from 30 E1 round-50 checkpoints: Full, FixedB-r8, and Two-r8, at targets one and two, across five seeds. Each receives two independently seeded Gaussian pulses at amplitudes $\alpha \in \{0.25, 1\}$ times its own per-round coordinate noise SD. This gives 120 probes. The pulse is a fresh perturbation of a saved state, not the replay or removal of an actual earlier training draw. One ordinary exposure round allows selected local vectors to respond before reset branches are constructed.

The unpulsed reference, retained-pulse branch, shared-reset branch, and local-reset branch then continue for ten rounds. A shared reset copies the reference shared state while retaining the pulsed branch’s local vectors. A local reset copies the reference local vectors while retaining perturbed shared parameters. A both-reset construction is an identity control. Branches share future client, negative-sampling, and coordinate-noise streams. Observations occur immediately after the pulse, after exposure, and at post-reset lags 0, 1, 4, and 10. All declared probes are retained.

9.2 Score decomposition and measurement

For reference (P, Q) and branch $(P + \Delta P, Q + \Delta Q)$,

$$\Delta S = P\Delta Q^T + \Delta P Q^T + \Delta P \Delta Q^T. \quad (11)$$

At a shared reset, $\Delta Q = 0$, so the remaining score difference is exactly $\Delta P Q^T$. Later differences may involve both states again. Squared norms of the three terms do not add to total squared norm without cross inner products; they are not additive causal percentages. The implementation records all terms and checks numerical closure in float64.

Measurements include score RMS across users and items, effective-matrix distance, top-10 disagreement on training-unseen candidates, and margin sign changes on 256 seeded random item pairs. The 942-user diagnostics do not use validation or test labels. “Exposed” users are those selected in the single exposure round; their identity is held fixed when reporting later outcomes. The normalized score ratio divides a probe’s delayed RMS by its own initial pulse RMS and is accompanied by the absolute value.

9.3 Initial findings and why a control was needed

At full pulse amplitude and lag ten after shared reset, Full has mean top-10 disagreement of 0.5212% at target one and 0.3206% at target two. FixedB-r8 has 0.3747% and 0.2038%. These modest effects show that restoring the shared item state does not automatically restore every local prediction. They do not establish an important relevance loss.

Raw-coordinate Two-r8 has much larger mean disagreement: 17.8546% and 9.3960%. Its target-one score-RMS/pulse ratio has mean approximately 911.2 but median only 1.065, indicating strong sensitivity to exceptional seeds. Treating this as typical persistent personalization damage would ignore both unstable scale and a deeper issue: the meaning of “the same noise” changes when factor coordinates are rebalanced.

9.4 Orthogonal alignment control

Equivalent factors can be rotated as $(A, B) \mapsto (AR, R^T B)$ for orthogonal R without changing Q . E2b aligns a branch to the current reference before each continuation round. It solves the

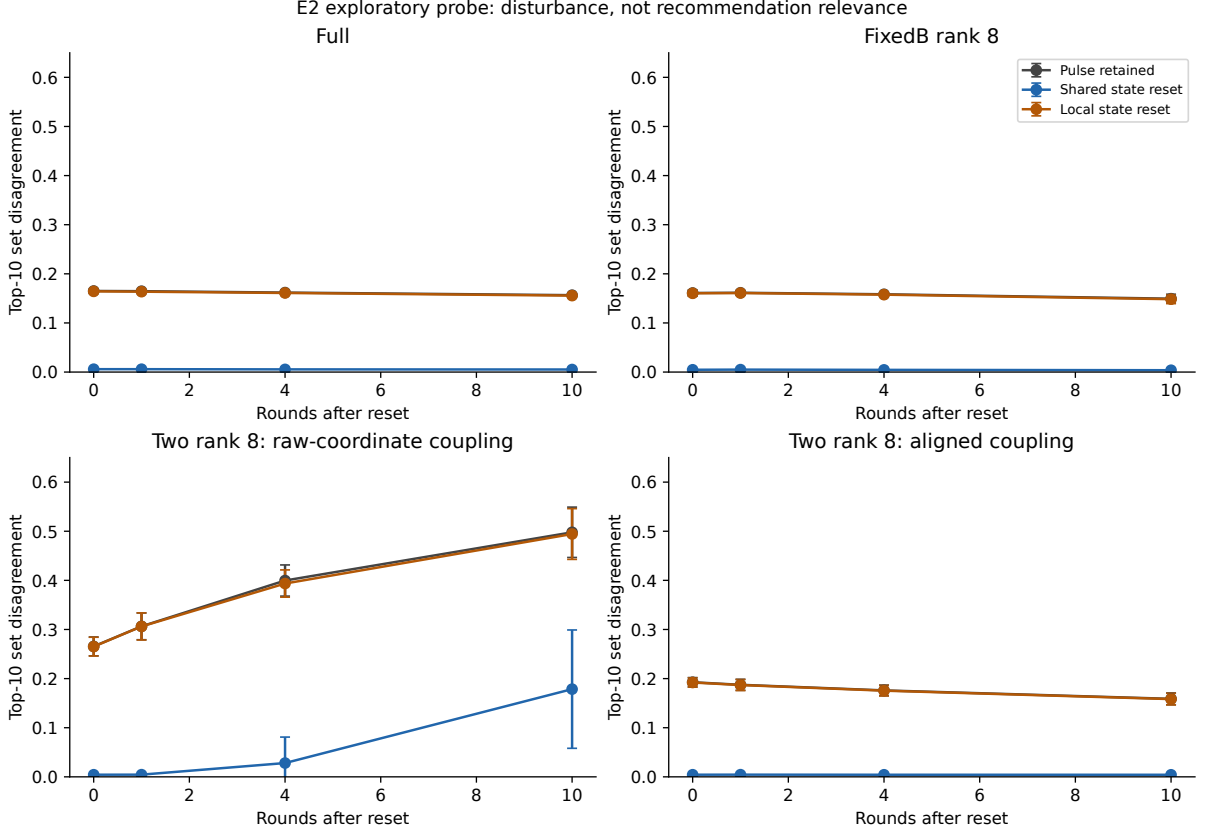


Figure 5: E2/E2b top-10 disagreement for retained pulse, shared reset, and local reset at starting target epsilon one and pulse amplitude one. Means and SDs use five checkpoint seeds after averaging two draws within seed. Candidate sets exclude training positives. The interventions access private states and measure prediction disagreement, not relevance or accuracy gains.

orthogonal Procrustes problem using the SVD of

$$A_{\text{branch}}^T A_{\text{ref}} + B_{\text{branch}} B_{\text{ref}}^T = U \Sigma V^T, \quad R = UV^T. \quad (12)$$

The transformed branch uses the same future coordinate draws in this aligned basis. Orthogonal transformations preserve isotropic Gaussian marginal laws in exact arithmetic, but alter the joint coupling between branches. Consequently a smaller paired distance does not imply an improved marginal training algorithm. Alignment here needs a counterfactual reference and is an oracle control, not a deployable intervention.

E2b completes 40 declared Two-r8 probes. Maximum measured relative product error from alignment is 3.9133×10^{-8} . At full amplitude, the lag-ten shared-reset disagreement falls to 0.4204% and 0.1083% at targets one and two. The large raw-coordinate distances are therefore highly coupling-sensitive. This does not identify alignment as a uniquely correct causal coupling: it demonstrates that the original interpretation was not invariant to a product-preserving representation choice.

9.5 Verification and limits

Six E2 and two E2b synthetic cases check zero-pulse equality, both-reset equality, RNG matching, reset identities, score closure, and product preservation. In real probes, both-reset states are constructed and checked at reset; the ten-round both-reset equality test belongs to the synthetic

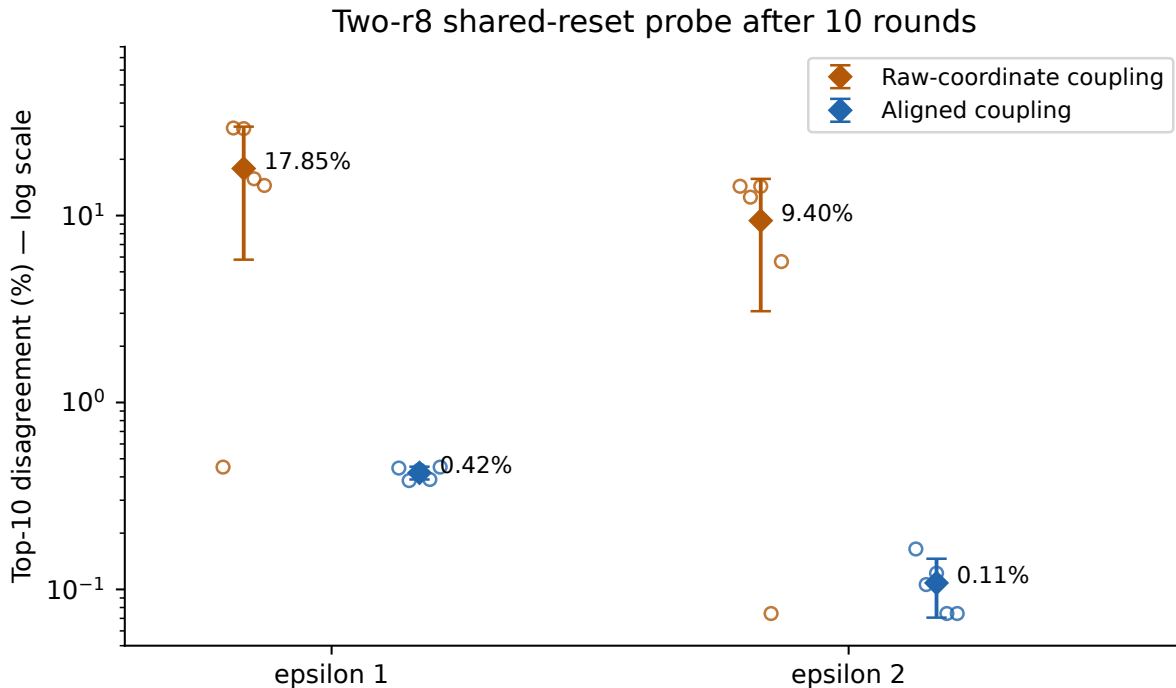


Figure 6: Two-r8 shared-reset disagreement at lag ten under raw and aligned coupling. Diamonds show means, whiskers seed SD, and circles all five seed values. The logarithmic axis reports percent top-10 set disagreement. Differences of 17.8546 versus 0.4204 percent at target one, and 9.3960 versus 0.1083 percent at target two, concern joint diagnostic coupling.

checks. This distinction avoids overstating the real experiment’s controls.

The aligned results leave modest local-state effects and an important measurement concern. E2b is an adaptive follow-up prompted by E2, disclosed as such. No held-out relevance evaluation or new privacy claim accompanies it. Continued releases would need additional composition beyond the starting epsilon. E3 tests whether the coupling phenomenon persists on a larger dataset and whether the balancing operation explains its size.

10 E3: larger-data replication and balancing controls

10.1 Transferred training and factorial diagnostic

E3 freezes its protocol on 3 October 2026 and trains 30 ML-1M models using the selected E1 settings without retuning: Full, FixedB-r8, and Two-r8; targets one and two; five seeds; 50 rounds. Local rate 5, two local epochs, $C = 1$, and $q = 0.1$ are unchanged. Full and FixedB retain server rate 1, while Two-r8 retains 0.5. The same noise multipliers give the same accountant bounds, but the larger qN reduces coordinate noise after normalization. Dataset size also changes the task, so differences in NDCG across datasets are not directly interpretable as a population-size effect.

The principal diagnostic crosses two datasets, two privacy labels, five seeds, two pulse draws, two noise couplings, and two continuation-balancing choices for Two-r8: 160 probes. Forty ML-1M Full/FixedB reference probes bring the total to 200. Pulse amplitude is one. Observer sets contain 128 users selected before outcomes are inspected. All users remain eligible for local updates. These streams and observer sets differ from E2, so their means need not reproduce

Table 8: Transferred MovieLens-1M validation NDCG@10.

Method	$\epsilon = 1$	$\epsilon = 2$
Full	0.019533 ± 0.004151	0.024624 ± 0.001777
FixedB r8	0.013661 ± 0.004574	0.016482 ± 0.004476
Two r8	0.015254 ± 0.005194	0.015440 ± 0.004567
DP popularity	0.026124 ± 0.000168	0.026161 ± 0.000104

Mean and seed SD, five seeds. Settings transfer without retuning. No ML-1M test scoring. Source: validation_summary.csv.

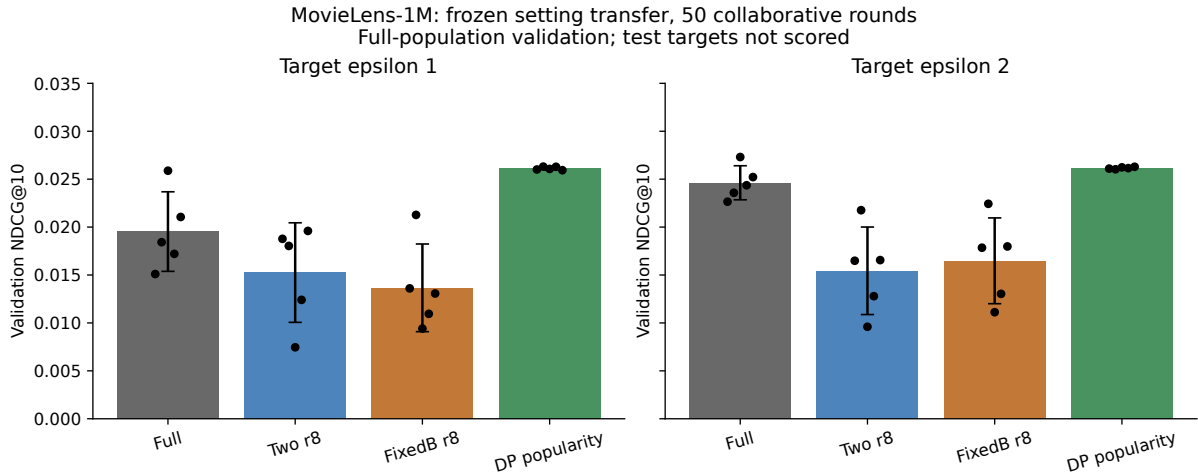


Figure 7: Full-population ML-1M validation NDCG@10 under frozen transferred settings. Bars and whiskers give five-seed means and SDs; dots show each seed. Collaborative models use 50 rounds and popularity uses one separately calibrated release. Point differences are descriptive. ML-1M test targets have not been scored.

E2’s numerical values exactly.

The no-balancing condition starts from the same balanced round-50 checkpoint and disables balancing only during the pulse/continuation experiment. It is not a model independently trained without balancing from initialization. Unlike merely changing a joint coupling, this changes the continuation algorithm itself. That distinction is central to its interpretation.

10.2 Validation utility

Table 8 reports full-population validation. All collaborative point means are below the separately calibrated private popularity reference. For example, at target two Full reaches 0.024624, FixedB-r8 0.016482, and Two-r8 0.015440, against popularity’s 0.026161. The fixed settings and short horizon may undertrain the larger problem; these numbers do not establish that tuned collaborative models cannot win. There is no paired superiority test or ML-1M test score in this stage.

10.3 Main diagnostic result

With SVD balancing, raw-coordinate coupling produces much larger shared-reset top-10 disagreement than aligned coupling on both datasets and privacy labels (table 9). On ML-1M, for example, target-one means are 5.2344% versus 0.0625%. Without continuation balancing, raw

Table 9: E3 shared-reset top-10 disagreement at lag ten (percent of set entries).

Dataset	ϵ	SVD/raw	SVD/aligned	None/raw	None/aligned
ML-100K	1	18.6641	0.4219	0.3438	0.3438
ML-100K	2	5.2031	0.0859	0.0781	0.0781
ML-1M	1	5.2344	0.0625	0.0703	0.0703
ML-1M	2	3.3438	0.0078	0.0156	0.0156

Two-r8; 128 observer users; two draws averaged within each of five seeds. None disables balancing only in the continuation. Source: endpoint_summary.csv.

and aligned churn are identical at saved precision for every seed. Small score RMS differences remain through float32 arithmetic.

These controls support a dependence on balancing-induced factor-basis changes. The initial large Two-r8 signal is not evidence for a general intrinsic failure of persistent local personalization. The result also does not mean balancing should simply be removed from a deployed optimizer: this stage does not compare independently trained balanced and unbalanced methods at equal privacy, utility, and convergence budgets.

The aligned diagnostic is small in ranking terms. ML-1M Full and FixedB at target one have mean shared-reset churn 0.1328% and 0.0859%, while aligned Two retains a score RMS about 3.6% of its initial pulse. In the 128-user observer sample, mean validation-NDCG changes under SVD/aligned Two are approximately 0.000040 and 0.000226 on ML-100K, and 0.000018 and 0 on ML-1M at targets one and two. These discrete, small- sample values do not establish equivalence, improvement, or absence of effects below their resolution.

10.4 Checks and research decision

All 30 training runs and 200 probes complete without failures. Twelve synthetic cases cover balanced-round identity, reset identities, zero pulses, and matching random streams. Maximum relative alignment product error is 3.9127×10^{-8} . The saved-evidence audit recomputes seed summaries from 2,800 observations and validation means from 181,050 saved user ranks, without scoring additional models.

The broad instability interpretation is rejected. A narrower empirical conclusion survives: representation-dependent coupling can dominate a particular paired noise-attribution diagnostic. Its scientific value is that it changes how the preceding experiment should be interpreted. Establishing a paper-level contribution would require showing that this issue materially changes conclusions or design decisions beyond the present simulator, in relation to existing gauge and alignment methods.

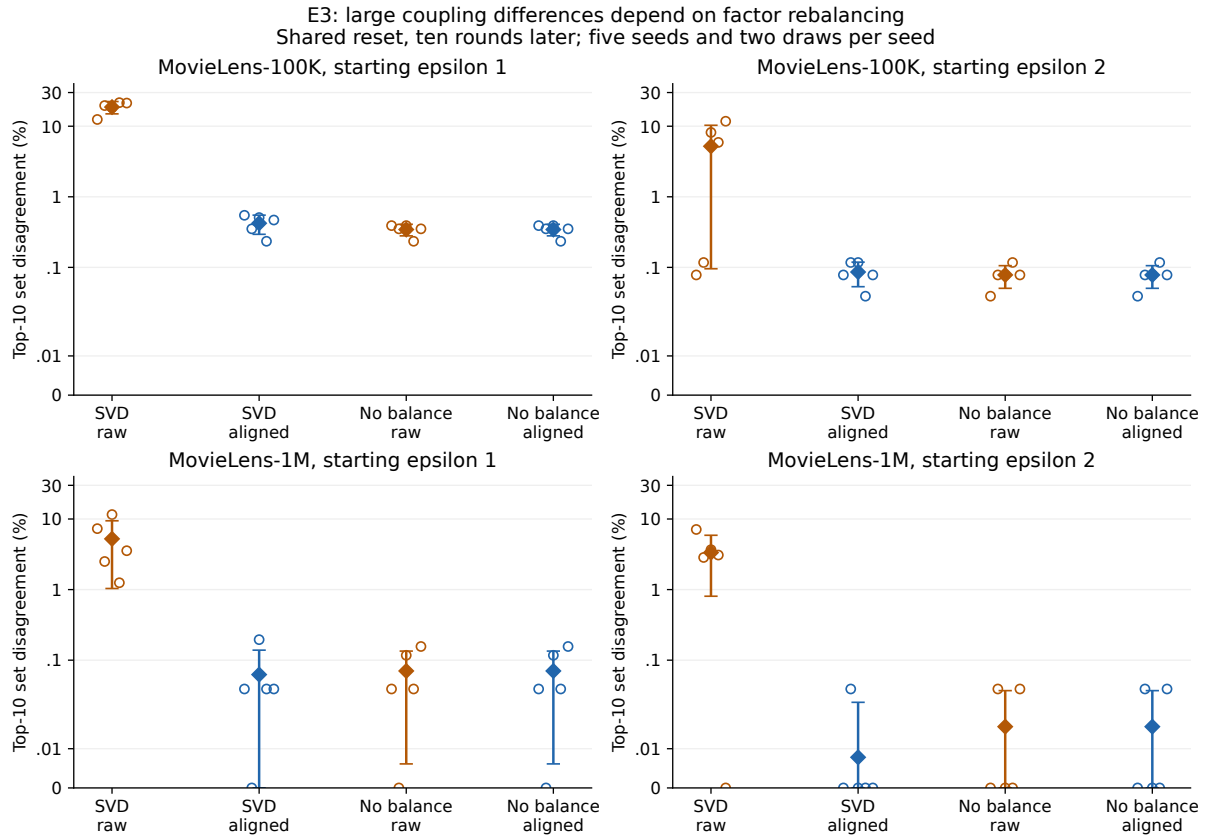


Figure 8: Two-r8 shared-reset disagreement after ten rounds on two datasets. Open circles show five seed values, diamonds means, and whiskers seed SD after averaging two draws. The symmetric-log axis is linear below 0.02 percentage points so exact zeros remain visible. Raw/aligned churn differs with SVD balancing and matches without it. Disabling balancing applies only to the continuation from the common balanced state.

11 E4: a conditional ranking-tail screen

11.1 Candidate explanation and established mathematics

The bilinear term’s small average energy in E1 does not logically guarantee accurate Gaussian tail probabilities. Ranking changes can depend on small margins, and rare-event probabilities may differ even when a variance component is modest. E4 tests this narrower possibility rather than assuming the energy result has settled all consequences of non-Gaussian noise.

For a fixed user vector p , item pair i, j , and signal factors, let $v = A_i - A_j$, $u = Bp$, $s_x = \sqrt{2}\tau$, and $s_y = \tau \|p\|$. The noisy margin has the conditional distribution

$$L = (v + s_x X)^T (u + s_y Y), \quad X, Y \stackrel{\text{ind}}{\sim} \mathcal{N}(0, I_r). \quad (13)$$

Set $a = v/s_x$, $b = u/s_y$, and $k = s_x^2 s_y^2$. With independent noncentral chi-square variables U, V having r degrees of freedom and noncentralities $\|a + b\|^2/2$, $\|a - b\|^2/2$,

$$L \stackrel{d}{=} \frac{\sqrt{k}}{2}(U - V). \quad (14)$$

This is an application of known distributional results [26]. For a positive clean margin $m = v^T u$, numerical integration evaluates $\Pr(U \leq V)$. It is compared with $\Phi(-m/\sqrt{V_{\text{lin}} + rk})$, where $V_{\text{lin}} = s_x^2 \|u\|^2 + s_y^2 \|v\|^2$, and with the linearized approximation $\Phi(-m/\sqrt{V_{\text{lin}}})$.

11.2 Synthetic checks and real-state selection

Six synthetic cases use 200,000 direct draws each to check moments and numerical integration, along with rank-one closed forms, symmetry, and rotation/positive-user-scale invariance. A declared rank-one example with $a = b = 4$ has bilinear variance share $1/33 = 3.03\%$. Its sign-flip probability is approximately 0.0000633405, versus 0.0026743853 for the variance-matched Gaussian, a factor of about 42.2. This construction shows why a tail check is worthwhile; it is not fitted evidence about the trained models.

The real-state screen uses 40 E1 ML-100K Two checkpoints (four ranks, two privacy labels, five seeds) and ten E3 ML-1M Two-r8 checkpoints. Sixteen preselected users per dataset and four training-unseen score-rank pairs, (10, 11), (1, 11), (10, 100), (1, 100), yield 3,200 records. The first two pair types are primary, giving 1,600 primary records. No model is trained and no held-out relevance label is read. The declared gate requires matched- Gaussian absolute error above 0.01 on at least 5% of primary records in both datasets.

11.3 Outcome and numerical scope

The maximum absolute matched-Gaussian error over all 3,200 records is 7.66859×10^{-6} ; the maximum linearized-Gaussian error is 7.26162×10^{-5} . Neither dataset has a primary record above the 0.01 threshold. The largest matched error is less than 0.000767 percentage points. The proposed advancement gate therefore fails. Group-mean pairwise bilinear variance shares range from approximately 0.0011% to 0.0750%, much smaller than the constructed example. These shares concern a particular user’s item-pair margin and are not the whole-matrix energy fractions reported in E1.

There are no tied, zero-user, or numerical-failure records. The largest reported quadrature error estimate is 2.64884×10^{-9} and the omitted tail mass is at most approximately 2×10^{-12} .

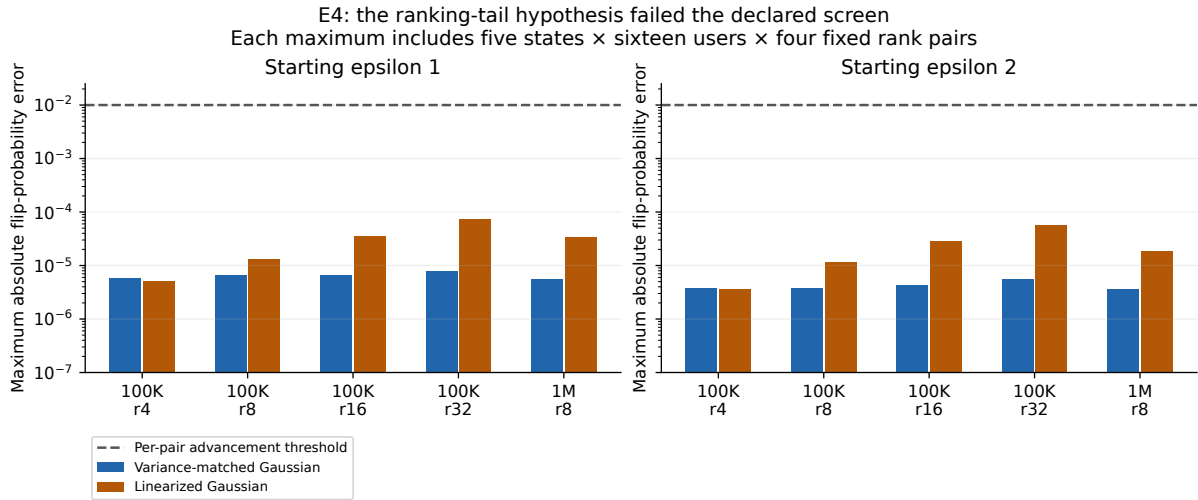


Figure 9: Maximum absolute flip-probability error for variance-matched and linearized Gaussian approximations against numerical evaluation of the known conditional law. Each bar covers 320 state-user-pair records. The logarithmic axis marks the 0.01 per-pair gate. Even maxima over all four pair types are far below the threshold. This is a one-step, label-free stability diagnostic, not a measurement of cumulative recommendation accuracy.

A square-root integration variable avoids the rank-one density singularity. These are numerical diagnostics, not a rigorous certified error bound for every possible parameter state. No expanded amplitudes, checkpoints, users, or pair types were added after inspecting the screen.

The result rejects practically consequential non-Gaussian approximation error for the sampled one-step round-50 states. It does not prove that all earlier iterates, all ranks, or multi-round training have Gaussian behavior. Nor does it refute noisy-score stability results with explicit distributional assumptions [25]. The correct decision is to retain the negative result and decline to build a new method around an effect that was negligible in the tested regime.

12 Discussion, limitations, and conclusions

12.1 Answers to the research questions

RQ1: representation and utility. Low rank reduces shared payload and can outperform the full private baseline in selected strong-privacy settings. The four positive adjusted historical comparisons support that qualified statement. Every nonprivate low-rank variant, however, underperforms the converged full model, and low rank does not dominate across privacy targets. The observed crossover combines capacity, optimization, clipping, and noise; it is not an isolated dimensional effect.

RQ2: fixed-factor effective noise. FixedB supplies a useful linear reference and smaller immediate perturbations in several diagnostics, but obtains no positive adjusted primary comparison against Two. The simple private popularity baseline exceeds all E1 collaborative point means at targets at most four. Smaller coordinate or effective-matrix noise alone is therefore an inadequate method-selection criterion in this benchmark.

RQ3: persistent local state. A perturbation to shared parameters can leave local user vectors changed even after the shared state is reset. The Full and FixedB probes show modest nonzero effects, as expected for coupled local/global dynamics. The evidence does not establish

a large general relevance loss caused by this pathway.

RQ4: validity of trajectory attribution. In rebalanced two-factor models, raw-coordinate common random numbers can produce much larger paired distances than an aligned coupling. The continuation-without-balancing control and ML-1M replication support a dependence on factor-basis changes. This finding changes the interpretation of E2: the raw trajectory distance cannot be read as a representation-invariant measure of an individual noise draw’s effect. It does not identify a unique causal coupling or a new private optimizer.

RQ5: ranking tails. The exact conditional margin law can differ substantially from a Gaussian in constructed regimes, but approximation error is negligible in the preselected real checkpoint screen. The non-Gaussian-tail candidate fails its practical advancement gate. The result narrows the explanation of these experiments without claiming Gaussianity throughout training.

12.2 What the combined evidence explains

The project demonstrates why an apparently intuitive story about private recommendation can fail. Reducing shared dimension lowers a coordinate-noise norm and payload, but also changes the learning geometry. Fixing a factor removes a bilinear term, yet that term’s measured energy can be small while the method still loses utility. Large personalized score disturbances can come from user scale, and large paired trajectory distances can arise from coordinate coupling rather than a comparably large marginal ranking effect.

The relevant sequence of measurements is therefore: the bounded client contribution, the effective shared signal and perturbation, the local user state, score margins, candidate rankings, and relevance. Communication should be measured both per exchange and over the full training schedule. These measurements complement the accountant, which constrains information release but does not predict usefulness. None is a substitute for a strong simple comparator such as properly private popularity.

The most distinctive completed empirical observation is the sensitivity of the personalized trajectory diagnostic to factor-basis coupling, supported by the larger-data and balancing controls. Its components overlap existing alignment and gauge methods. This report therefore treats it as a supported diagnostic result whose paper-level novelty remains unconfirmed, not as an established breakthrough or a new privacy theorem.

12.3 Threats to validity

Selection and benchmark exposure. ML-100K has repeated historical exposure, including the disclosed accidental regression-test evaluations. Later freezes improve local experiment discipline but do not undo accumulated development knowledge. E2b is an adaptive follow-up, and E3/E4 questions were informed by earlier results. ML-1M validation is now exposed; its test remains available for a future genuinely frozen comparison.

Optimization and tuning. Historical B4 has more tuning than B2. E1 equalizes a finite candidate count, not the attainable optimum of every method, and some selected rates are at the boundary. Nonprivate convergence experiments and 50-round private runs have different budgets. ML-1M transfer uses no retuning. These choices make the recorded comparisons reproducible but do not prove method-independent limits on privacy or rank.

Statistical interpretation. User-bootstrap intervals condition on retained training runs. Five seeds provide limited information about rare training instabilities. E2/E3 probes are correlated within checkpoints and are summarized descriptively. The small observer subset in E3 limits resolution of validation differences. Activity, cold-item, and personalization ablations are exploratory rather than multiplicity-adjusted subgroup discoveries.

Privacy and systems scope. The simulator assumes only noised aggregate visibility and implements no secure aggregation. Public population/catalog assumptions, nonprivate selection, internal checkpoints, and diagnostic releases fall outside the per-run guarantee. The modeled communication excludes protocol overhead and client availability costs. It does not evaluate a deployed mobile system.

Generalization and numerical scope. Both benchmarks are MovieLens variants, not independent services. Per-user chronological splitting is not a global future-data prediction experiment. There are very few cold targets. Effective-noise formulas condition on a single state, while E4 samples a finite collection of round-50 margins. Large finite local vectors and float32 update arithmetic require careful numerical interpretation. The report makes no claim of universal bitwise reproducibility across environments.

12.4 Future work justified by the evidence

The first priority is a faithful published private ALS or federated-MF comparator. Such a comparison requires matching the privacy unit, adjacency, trust model, feedback objective, candidate filter, and communication definition. Perturbing sufficient statistics rather than BPR updates may change the relevant signal-to-noise regime. The literature/implementation audit is complete enough to specify this next comparison, but no such baseline result is included as completed work.

The second priority is metadata-only local personalization, followed by a small private collaborative residual. Public genres or other permitted item features could supply useful structure before any private collaborative update is spent. A meaningful extension would test incremental residual value against that public-only control and private popularity under an explicit total communication cap. Cold items and low-activity users are natural targets, provided the dataset supplies enough examples for reliable analysis. The underlying public-feature idea already has prior work.

A third possibility is to test whether coupling-aware diagnostics change a consequential scientific conclusion outside the present pilot. The target conclusion and decision criterion should be frozen before running more probes. Merely repeating the same alignment contrast at additional amplitudes would add volume without necessarily establishing novelty. A useful method would also need to avoid counterfactual access to private reference states and receive fresh privacy accounting.

12.5 Conclusion

This project began with low-dimensional sharing as a possible route to better private federated recommendation and developed into a systematic study of how parameter noise becomes prediction disturbance. The final record includes a reproducible benchmark, controlled fixed-factor and private-popularity comparisons, conditional noise analysis, pulse/reset experiments,

a larger-data replication, and a rejected ranking-tail explanation. The evidence supports conditional low-rank gains and a specific limitation of factor-coordinate trajectory measurements, while rejecting broader claims that fixed factors, smaller coordinate norms, or bilinear tails alone solve the observed utility problem.

The principal outcome is a more precise experimental foundation for the next research decision. It preserves both favorable and unfavorable results, defines the scope of privacy and uncertainty, and identifies stronger comparators and public-information controls needed before claiming a new recommendation method. That foundation, rather than an unsupported novelty claim, is the completed contribution of this undergraduate project.

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A Mathematical derivations and diagnostic identities

A.1 Conditional effective-matrix moments

Let A, B, P be fixed conditional states after the deterministic signal part of a round. Assume independent entries of $Z_A \in \mathbb{R}^{M \times r}$ and $Z_B \in \mathbb{R}^{r \times d}$ have mean zero and variance τ^2 , with the two matrices independent. In the implemented mechanism they are Gaussian. Write $D_1 = AZ_B$, $D_2 = Z_A B$, and $D_3 = Z_A Z_B$.

Using $\mathbb{E}[Z_B Z_B^T] = d\tau^2 I_r$ gives

$$\mathbb{E} \|D_1\|_F^2 = d\tau^2 \|A\|_F^2, \quad \mathbb{E} \|D_2\|_F^2 = M\tau^2 \|B\|_F^2. \quad (15)$$

Every entry of $Z_A Z_B$ is a sum of r products of independent centered variables, with variance $r\tau^4$. Summing over its Md entries gives $\mathbb{E} \|D_3\|_F^2 = Mdr\tau^4$. Each cross expectation vanishes by conditioning on one centered noise matrix. For example, conditional on Z_B , the inner product $\langle AZ_B, Z_A Z_B \rangle_F$ is linear in Z_A and has expectation zero. Expanding the squared norm proves equation (9); mutual independence of all three products is unnecessary.

A.2 Score moments and pairwise margins

For the first score term, $PD_1^T = PZ_B^T A^T$, the identity $\mathbb{E}[Z_B^T A^T A Z_B] = \tau^2 \|A\|_F^2 I_d$ yields

$$\mathbb{E} \|PD_1^T\|_F^2 = \tau^2 \|A\|_F^2 \|P\|_F^2. \quad (16)$$

For $PD_2^T = PB^T Z_A^T$ the expectation is $M\tau^2 \|PB^T\|_F^2$. Conditioning successively on both noises gives $Mr\tau^4 \|P\|_F^2$ for PD_3^T . Cross expectations again vanish, giving equation (10).

For distinct items i, j and fixed user p , the noisy difference between their scores has a centered shock with variance

$$V_{ij} = 2\tau^2 \|Bp\|^2 + \tau^2 \|A_i - A_j\|^2 \|p\|^2 + 2r\tau^4 \|p\|^2. \quad (17)$$

The factor two arises because the two independent rows of Z_A are subtracted. Shared Z_B affects both items through their factor difference. The full shock is not generally Gaussian. For clean nonzero margin m , Chebyshev's inequality supplies the weak bound

$$\Pr(\text{sign flip}) \leq \Pr(|\text{shock}| \geq |m|) \leq \min(1, V_{ij}/m^2). \quad (18)$$

A variance alone does not justify a Gaussian tail formula for Two.

For FixedB the shock is $(Z_{A,i} - Z_{A,j})Bp$, exactly Gaussian with variance $2\tau^2 \|Bp\|^2$. Its fixed-margin sign-flip probability is $\Phi(-|m|/(\sqrt{2}\tau \|Bp\|))$ when the denominator is positive. This formula conditions on the current factors and user state; it is not an expression for full-catalog NDCG or for accumulated training loss.

A.3 Scale changes, balancing, and clipping

The transformation $(A, B) \mapsto (cA, B/c)$ preserves Q for positive c . The linear portion of expected matrix-noise energy becomes

$$\tau^2 \left(\frac{M \|B\|_F^2}{c^2} + dc^2 \|A\|_F^2 \right). \quad (19)$$

Differentiating gives the minimizer $c^4 = M \|B\|_F^2 / (d \|A\|_F^2)$ when both factors are nonzero. For scores, the corresponding minimizer satisfies $c^4 = M \|PB^T\|_F^2 / (\|A\|_F^2 \|P\|_F^2)$. An equal-factor-norm convention need not minimize either rectangular or user-weighted expression. The bilinear τ^4 term is unchanged by this scalar gauge transformation.

These conditional optima are not a training prescription. Changing factor scale also changes client gradients and which updates are clipped. A rule chosen from private P would require a privacy treatment of that information. Balancing after the noisy update preserves its already-noisy AB and cannot remove the realized effective shock. The formulas explain an interpretation problem, not a free improvement of the privacy mechanism.

Separate factor averaging has an additional distinction. With client means $\bar{A}, \bar{B}$,

$$\frac{1}{K} \sum_{u=1}^K A_u B_u = \bar{A} \bar{B} + \frac{1}{K} \sum_{u=1}^K (A_u - \bar{A})(B_u - \bar{B}). \quad (20)$$

Thus averaging factors and averaging products generally differ by a cross-client covariance term. FixedB removes this discrepancy, but E1 does not independently intervene on it. It cannot be isolated as the reason for a utility difference from the existing comparisons.

A.4 FixedB score capacity and regularization

If $BB^T = I_r$, define $v_u = Bp_u$. The scores are $A_i v_u$, with a score matrix of rank at most r . Conversely any desired $v_u \in \mathbb{R}^r$ has preimage $p_u = B^T v_u$, and every other preimage adds a vector in the null space of B . Orthogonality implies

$$\min_{p: Bp=v} \|p\|^2 = \|v\|^2, \quad \|A_i B\|^2 = \|A_i\|^2. \quad (21)$$

Projecting the local gradient through B therefore gives ordinary rank- r BPR dynamics for the score-relevant vector, under the same effective-row penalty, up to floating-point behavior. A null-space component does not change scores. FixedB does not encode public item semantics; it defines a public coordinate map for private collaborative factors.

A.5 A product-preserving coupling null example

The pairs (A, B) and $(-A, -B)$ represent exactly the same matrix. Applying identical raw-coordinate Gaussian draws to both gives noisy products whose difference is

$$(A + Z_A)(B + Z_B) - (-A + Z_A)(-B + Z_B) = 2(AZ_B + Z_A B). \quad (22)$$

The expected squared distance is $4\tau^2(d \|A\|_F^2 + M \|B\|_F^2)$, although the two noisy products have the same marginal distribution. Coupling the second draw as $(-Z_A, -Z_B)$ instead makes the products identical. A retained 10,000-draw synthetic check gives measured energy 6.341142 against theoretical 6.345313, with Monte Carlo SE 0.012443; aligned distance is exactly zero. This elementary example makes the coupling issue explicit without claiming a new theorem or an accuracy gain.

A.6 Exact conditional margin law and numerical evaluation

Let $\tilde{X} = X + a$, $\tilde{Y} = Y + b$ in equation (13). Then $(\tilde{X} + \tilde{Y})/\sqrt{2}$ and $(\tilde{X} - \tilde{Y})/\sqrt{2}$ are independent normal vectors with identity covariance and means $(a + b)/\sqrt{2}$ and $(a - b)/\sqrt{2}$. The identity

$\tilde{X}^T \tilde{Y} = (\|\tilde{X} + \tilde{Y}\|^2 - \|\tilde{X} - \tilde{Y}\|^2)/4$ proves equation (14). The first three cumulants are

$$\mathbb{E}L = m, \quad \text{Var}(L) = V_{\text{lin}} + rk, \quad \kappa_3(L) = 6km. \quad (23)$$

For positive m , the sign probability can be evaluated as $\int_0^\infty F_U(v)f_V(v) dv$. The implementation controls omitted tails, uses a square-root variable near zero, and compares against direct Monte Carlo in synthetic cases. Positive rescaling of p multiplies both margin and shock by the same positive amount, leaving sign-flip probability unchanged. The real-state calculation normalizes this scale for numerical stability; it does not change the ranking question.

A.7 Analytic Gaussian calibration for popularity

Under add/remove adjacency, equation (6) changes the unnoised sum by a vector of norm at most $\sqrt{20}$. For noise SD equal to sensitivity times σ_{pop} , the analytic Gaussian condition is

$$\delta = \Phi\left(\frac{1}{2\sigma_{\text{pop}}} - \epsilon\sigma_{\text{pop}}\right) - e^\epsilon \Phi\left(-\frac{1}{2\sigma_{\text{pop}}} - \epsilon\sigma_{\text{pop}}\right). \quad (24)$$

Solving numerically yields multipliers approximately 0.600229, 1.081162, 1.993812, and 3.730632 for targets 8, 4, 2, and 1. This is one release with $q = 1$, not 50 compositions of a subsampled update. Releasing all noise-seed versions would require additional accounting. The derivation uses the established analytic Gaussian mechanism [8].

B Implementation details and algorithm record

B.1 Client update and shared aggregation

Each federated local epoch is a full-batch SGD step over that client’s training positives, using one fresh uniform negative per positive. This normalizes the loss within a user. Repeated item indices in positive and negative samples accumulate their gradients. Only touched item rows receive local loss gradients, but private aggregation subsequently adds noise to every shared coordinate. Item rows not touched by any user therefore can still change in a private round.

1. Sample each eligible user independently with public inclusion probability q . Preserve the selected cohort for all shared and local computations.
2. For each selected user, copy the common shared state and retrieve its local vector. Repeat $E = 2$ times: draw negatives from the training-only mask, compute the within-user mean BPR loss and explicit regularizer, and update both the local vector and the client’s shared copy.
3. Persist the new local vector. Form the difference between the client’s shared copy and the round’s starting state. For a private run, jointly clip that whole difference to norm C .
4. Sum the contributions. In private runs, add an independent Gaussian vector with SD σC , even when no user was selected. Divide by fixed qN and multiply by η_s . Nonprivate B1 instead averages over realized participants and skips an empty round.
5. Update the server’s shared state. Where specified, rebalance the two factors without changing their product. Record communication and diagnostics at the protocol’s declared observation points.

This procedure describes the simulator’s semantics. It does not specify a network protocol, secure-aggregation implementation, or availability model.

B.2 Final parameter register

Table 10: Final retained configurations and stopping rules.

Model	Settings
B0 / B0-UW	$d = 64$, initialization SD 0.01; Adam 10^{-3} ; L2 10^{-2} ; batch 1,024; maximum 1,000 epochs; patience 100. Uniform interactions / uniform users then positives, respectively.
B1	$q = 0.1$, $E = 2$, local rate 5, L2 10^{-5} , server rate 1; validation every ten rounds; patience 100 validations; maximum 8,000 rounds.
B2	B1 local settings; $T = 50$, $C = 1$, fixed qN , final-round checkpoint, PRV-calibrated noise and $\delta = 10^{-5}$.
B3	Ranks 4/8/16/32; local rates 0.625/1.25/1.25/0.625; effective-row L2; QR/core-SVD balancing; B1 stopping with the disclosed r32/seed123 ceiling exception.
B4	$T = 50$, $C = 1$, same private calibration as B2; r4 local/server 2.5/1; r8/r16/r32 local/server 5/0.5; joint factor clipping and full-coordinate noise.
E1 Full	Local/server 5/1, six levels including no-clipping/no-noise and clipping-only controls; same final iterative calibration.
E1 Two	Local 5; server 0.5 for r4/r8/r16 and 1 for r32; initial effective matrix matched to FixedB; both factors trained and balanced.
E1 FixedB	Local/server 5/1; orthonormal public basis seed 314159; private A only; 64-dimensional local vectors; no semantic metadata.
E1 popularity	Binary distinct-training-item contributions; norm bound $\sqrt{20}$; full population; one analytic-Gaussian release.
E3 transfer	E1 Full, FixedB-r8, and Two-r8 settings without retuning; two privacy targets, five seeds, 50 rounds, ML-1M normalization.

B.3 Diagnostic random streams and identity checks

E2 pulse streams use the tuple (training seed, 6702, pulse replicate). Future client, negative, and Gaussian streams use identifiers 6703, 6704, and 6705 respectively. Common streams are retained across branches and pulse amplitudes. E3 uses separately declared dataset-coded streams with identifiers 6710–6713 and observer selection identifier 6708. E4 uses its own fixed observer selection. Exact seed tuples are part of the frozen protocols and saved records, so a report rerender does not generate new diagnostic observations.

Synthetic checks compare zero-pulse branches with their references and both-reset branches over continued rounds. Shared resets enforce identical Q ; local resets enforce identical P . The score decomposition includes all cross inner products. Alignment checks measure relative product error instead of assuming exact equality in float32. E4 checks moment formulas within Monte Carlo uncertainty and audits quadrature diagnostics separately. These checks support

the implemented experiments; they do not prove an unrestricted mathematical correctness claim for every numerical state.

B.4 Numerical environment and reproducibility precision

The pinned research environment records NumPy 2.2.6, pandas 2.3.3, SciPy 1.15.3, PyTorch 2.14.0+cpu, Opacus 1.6.0, matplotlib 3.10.9, and the remaining versions in the curated requirements file. Opacus is used for accountants, not a PrivacyEngine wrapping these models. Training and diagnostic observer precision are distinguished: model updates retain float32, while relevant diagnostic norms and closures use float64. E1 enforces one numerical worker thread; older centralized runs used their recorded CPU/thread configuration.

A fresh-environment historical B0 reproduction was identical to reported metric precision but not byte-identical for seed 42. Relative weight difference was approximately 7.8×10^{-6} , and one validation rank changed from 596 to 595 without changing reported six-decimal metrics. A floating-point kernel explanation was plausible but unverified. Accordingly the report promises recorded seeds, environments, artifacts, and numerical audits, not universal bitwise identity on different systems.

B.5 Communication ledger

Table 11: Modeled dense tensor communication for retained historical methods.

Model	Bytes/client/round	Rounds to best	GB to best	GB complete
B1	861,184	1158.0	93.911	174.946
B3 r4	55,872	2984.0	15.704	20.971
B3 r8	111,744	3064.0	32.263	42.790
B3 r16	223,488	3676.0	77.388	98.415
B3 r32	446,976	4284.0	180.329	222.297
B2	861,184	50.0	4.026	4.026
B4 r4	55,872	50.0	0.261	0.261
B4 r8	111,744	50.0	0.522	0.522
B4 r16	223,488	50.0	1.045	1.045
B4 r32	446,976	50.0	2.090	2.090

Decimal GB; realized-participation totals averaged over seeds. Headers, secure aggregation, compression, and device costs excluded. Source: communication.csv.

Totals in table 11 distinguish the selected checkpoint from the complete run, including patience rounds. B2/B4 have identical selected and completed horizons; B1/B3 do not. The historical column comparing every model to B1 is not used to claim a like-for-like private communication advantage: B2/B4 should be compared at their matched 50-round horizon. Decimal gigabytes and dense float32 counting are used throughout.

C Material revisions and hypothesis decisions

C.1 Why the project history belongs in the report

The final tables are not the first outputs produced by the project. Stopping rules, stability gates, and proposed explanations were revised as new evidence became available. The purpose of this appendix is to disclose the revisions that affect interpretation, without presenting every abandoned grid cell as another independent result. Complete chronological commands, logs, superseded outputs, and freezes remain in the research archive.

Table 12: Material research iterations and their consequences.

Stage	What changed or was learned	Consequence for interpretation
Phase 0, 28 Sep.	Positive threshold, per-user chronology, hashed timestamp ties, candidate filtering, and raw/split hashes were fixed.	A common full-ranking protocol replaces inconsistent or sampled-candidate comparisons.
B0, 28–29 Sep.	Short patience and epoch caps could stop a still-improving configuration; validation-only convergence checks revised selection.	Earlier B0 numbers remain historical, not the final common-protocol reference.
B0/B1, 2 Oct.	A common longer stopping protocol and centralized B0-UW control were added.	User weighting partly explains the apparent federation difference; overall paired intervals include zero.
B2, 2 Oct.	T1000/C1.5/server2 was replaced after validation studies by T50/C1/server1.	Three design choices changed; a single-factor causal story about horizon is not supported.
Test-path audit	Nine accidental two-round B2 real-data test evaluations occurred in regression checks; a cache-key issue was corrected.	No selection used those outputs, but ML-100K is not pristine. The incident is retained.
B3	Initial rates caused divergence; a rank-32 convergence/stability gate required bounded correction before test scoring.	Report final rates and the one declared 16,000-round diagnostic ceiling. Do not silently drop failed attempts.
B4	Inherited long-horizon rates undertrained at 50 rounds; expanded searches and a bounded fallback selected finite settings.	Historical B4 has a larger tuning budget than B2, limiting a dimension-only comparison.
E1, 2 Oct.	A new equal-four-candidate extension matched factorized initializations and added DP popularity and clipping controls.	E1 is separately named and frozen; it does not overwrite B4. Eleven search failures remain counted.

Table 12 (continued)

Stage	What changed or was learned	Consequence for interpretation
E1 diagnostics	Float32 local norm summaries overflowed for a finite trajectory; float64 analysis corrected the summary.	Training, checkpoint contents, and test ranks are unchanged. Extreme finite results stay visible.
E1 verification	Live test stdout written under a guarded results tree interfered with integrity fixtures. A rerun wrote to a temporary location.	The recorded passing run has 203 tests and 26 warnings; both logs are retained. The failure was a harness/output-location issue.
E2, 3 Oct.	120 declared pulse/reset probes suggested local memory and large Two-r8 distances.	Exploratory diagnosis, no relevance scoring or new DP claim.
E2b, 3 Oct.	An alignment control was declared after E2's large two-factor distances.	Adaptive follow-up sharply narrows the original interpretation.
E3, 3 Oct.	30 ML-1M training runs and 200 raw/aligned, balanced/unbalanced probes completed.	The large coupling effect depends on balancing; ML-1M validation is exposed, while its test is unscored.
E4, 3 Oct.	50 states and 3,200 prescribed pairs tested a known non-Gaussian margin law.	The practical-error gate failed. No post hoc expansion was used to rescue the candidate.
Report, 5 Oct.	Saved evidence was organized into LaTeX narrative, vector figures, generated tables, and supplementary material.	No new training, model selection, or test scoring was needed for report preparation.

C.2 Final hypothesis ledger

Fewer shared noisy coordinates reliably improve recommendation.

Rejected as a general statement. Four favorable historical private contrasts coexist with weak-privacy losses, nonprivate losses, and a strong private popularity comparator.

Fixing one public factor is a sufficient new solution.

Unsupported by E1 utility, and already preceded by fixed-factor literature. The implemented method is a controlled baseline.

The factor-noise product dominates the observed failures.

Not supported by its small measured matrix-energy share. E4 additionally rejects large Gaussian tail-risk approximation error in the prescribed states.

Noise causes a large general persistent-personalization failure.

The broad reading of raw Two-r8 distances is rejected by alignment and no-balancing controls. Modest local-state effects remain.

Factor-basis changes can dominate a paired noise-attribution measurement.

Supported within the declared simulator, counterfactual, datasets, and observation rules.

Its external consequence and paper novelty remain open.

Public metadata or private ALS has already solved the remaining problem here.

Not tested. Literature review and comparator planning must not be presented as completed method implementations or experimental results.

C.3 What was deliberately excluded from the submission tables

Superseded sweeps, bulk checkpoints, raw user records, and every intermediate log are not copied into the submission package. Their omission serves traceability: the formal tables should have one unambiguous retained source. Failures and revisions that influence conclusions are summarized above and in the source inventory. Unfavorable final results, all declared primary contrasts, and secondary diagnostic families are retained in the supplement. The source archive remains the complete record of the research process.

D Artifact provenance and reproduction guide

D.1 Separation of evidence, report source, and generated output

The sibling `fedrec-dp` repository contains experiment implementations, configuration files, chronological research notes, raw run logs, and frozen checkpoints. The `CS-670` directory curates material relevant to the course report. Within it, `submission` contains the portable LaTeX project: main report, supplement, chapter files, vector figures, bibliography, copied numerical evidence, and a table-generation script. The earlier `report/UGP_REPORT_DRAFT.md` is a working narrative record; this LaTeX document is the formal submission version.

Copied source records have explicit source paths and SHA-256 hashes. The report does not obtain its numbers by re-evaluating saved models. Tables are formatted from retained CSVs, and the generator records hashes of every input. Figures are the nine existing audited vector exports. Their data sources, SD conventions, and transformations are stated in captions. Extra digits in the supplement support traceability and should not be interpreted as additional inferential precision.

D.2 Data fingerprints

The ML-100K raw ratings SHA-256 is

06416e597f82b7342361e41163890c81036900f418ad91315590814211dca490.

Its frozen split fingerprint is

6faed6fc3d47b5b6fa638adfeea83cd7409d50c39fa01f85c10379d0daef6989.

The ML-1M split fingerprint is

e68ef765cc661f5db1864f9f8b2843bee70c7ad261c116421d2c7102ec71bd54.

Its downloaded archive MD5 is `c4d9eecfca2ab87c1945afe126590906`. The E3 provenance file contains the additional archive SHA-256, individual raw/processed hashes, and selected observer IDs. Raw datasets are research inputs outside the portable writing package; the GroupLens distribution terms apply.

D.3 Evidence checks

The retained bundle audit verifies 3,932 protected historical artifacts, the E1 manifest and snapshots, copied-source provenance, frozen diagnostic sources, and checkpoint hashes. It recomputes pilot summaries, checks reset and score-energy identities, and verifies E3/E4 summaries against original observations when the sibling archive is available. E1 records a completed 203-test suite with 26 warnings. E2/E2b, E3, and E4 add their own declared synthetic checks; these are not described as a new run of that full suite.

Useful archive-relative entry points are

- `src/data.py` and `src/evaluate.py`: canonical split and ranking;
- `src/federated.py` and `src/effective_noise.py`: federated and E1 mechanisms;
- `experiments/run_effective_noise.py`: controlled E1 runs;
- `experiments/probe_noise_memory.py` and `experiments/probe_noise_memory_aligned.py`: E2/E2b;
- `experiments/run_noise_replication.py`: E3;
- `experiments/probe_ranking_tails.py`: E4;
- `experiments/summarize_noise_followup.py`: saved-evidence E3/E4 analysis.

The full research log records exact original commands. Reproduction should use a separate output root; frozen historical files must not be overwritten.

D.4 Rebuilding the submission

From the submission directory, run the table generator with a Python environment containing pandas, then compile `main.tex` and `supplement.tex` with a standard LaTeX distribution. The supplied Makefile supports `latexmk`; the build script also supports Tectonic. BibTeX uses the included `references.bib`. The source archive includes all figure PDFs and table fragments, so compiling alone does not require rerunning Python or obtaining the research datasets. No shell-escape feature or remote figure URL is required.

Author, student ID, supervisor, department, and institution are optional fields in `metadata.tex`. They are omitted while unspecified; the course code and report date are supplied. Institutional certification, signatures, and declarations are not fabricated. A department-specific cover page can be substituted without changing the scientific content.

D.5 Supplementary material and source map

The companion supplement provides all 34 historical five-seed utility rows, all 59 E1 utility rows, all 53 historical and 117 E1 paired contrasts, complete E1 geometry summaries, activity/cold-target tables, personalization ablations, diagnostic endpoints, privacy calibration, and rank selections. Machine-readable CSVs retain full saved precision. Source hashes and an evidence map connect each table family to the corresponding retained experiment, preventing a report editor from inadvertently substituting an older, superseded result with a similar filename.